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A Complete No-Brainer: ReRAM for Neuromorphic Computing

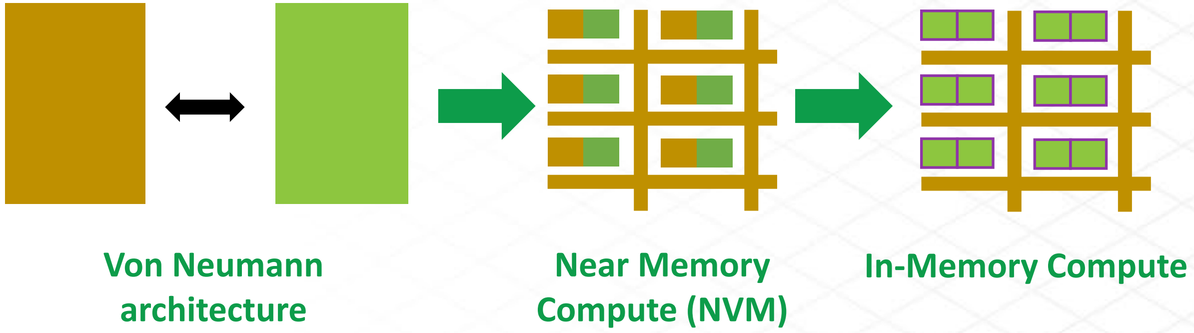
By Giuseppe Piccolboni, R&D Engineer

May 2024

Select Slides

imw 2024

Non-von Neumann Computing



- Non-von Neumann systems to bring memory closer to computation for better efficiency and lower power consumption
- In memory computing (IMC): computation is done directly within the memory
- Change of paradigm: new devices are required

Outline

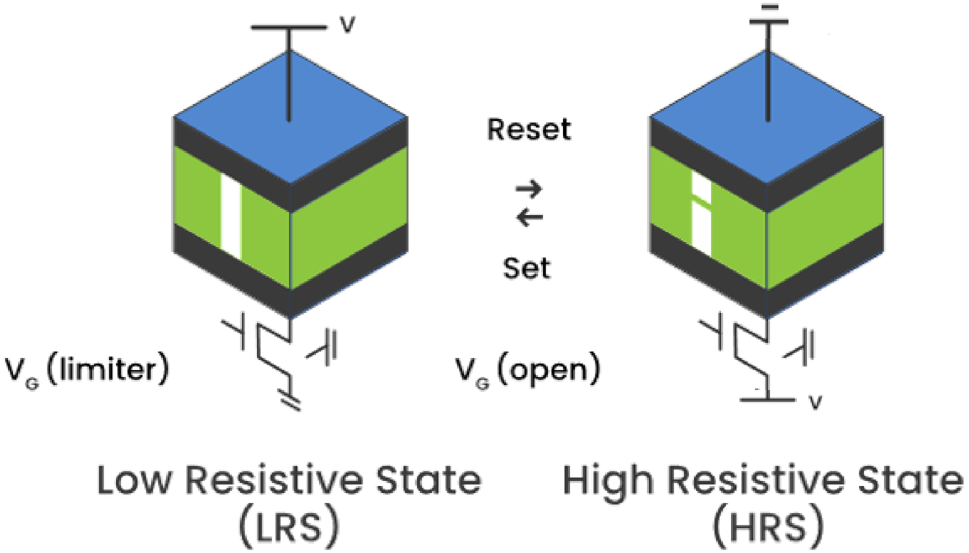
- **ReRAM characteristics**
  - Endurance, retention, …
  - State-of-the-art
- **ReRAM device challenges for Neuromorphic circuits**
  - Multi-level operations
  - Relaxation and fluctuation mitigation
- **Roadmap for ReRAM in AI circuits**
  - Embedded memory for synaptic weight storage
  - In memory computing
  - New concepts
- **Conclusions**

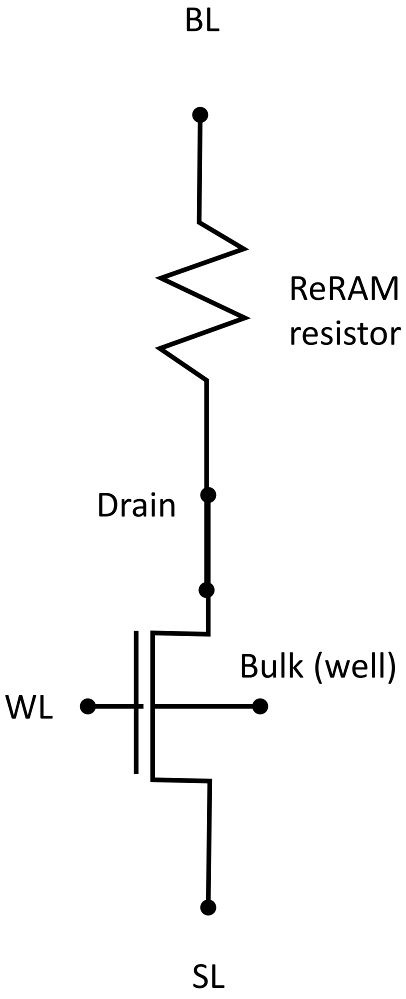
ReRAM Basic Operation

- **ReRAM is based on oxygen vacancies filament (OxRAM)**
  - By applying different voltage levels on the resistive layer, a filament is created or dissolved
    - RESET (Erase) – Partial dissolution of the Conductive Filament: LRS –> HRS
    - SET (Program) – Recreation of the Conductive Filament: HRS->LRS
  - **Data retained within the stack is resilient to many environmental conditions**

Low Power Consumption

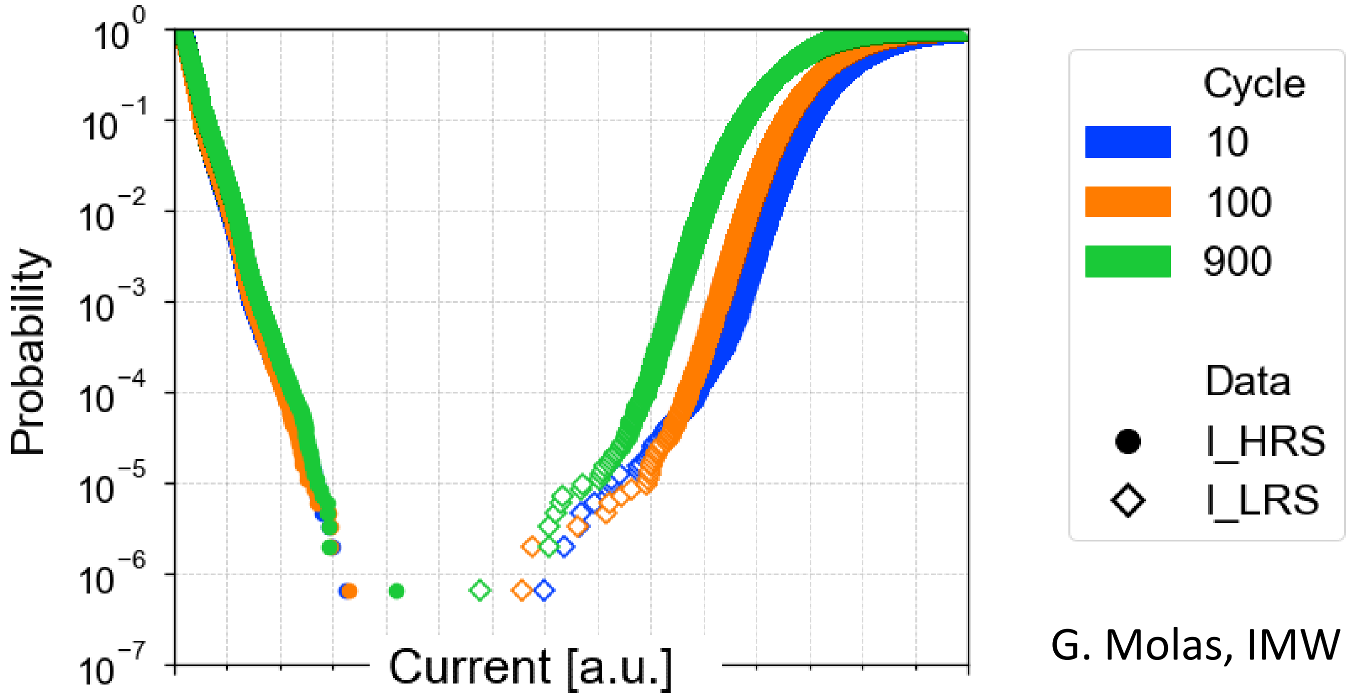
- ✓ Low read voltage <1V
- ✓ Low write voltage <3V
- ✓ No erase before write
- ✓ Low currents
- ✓ Zero standby power
- ✓ Fast operation



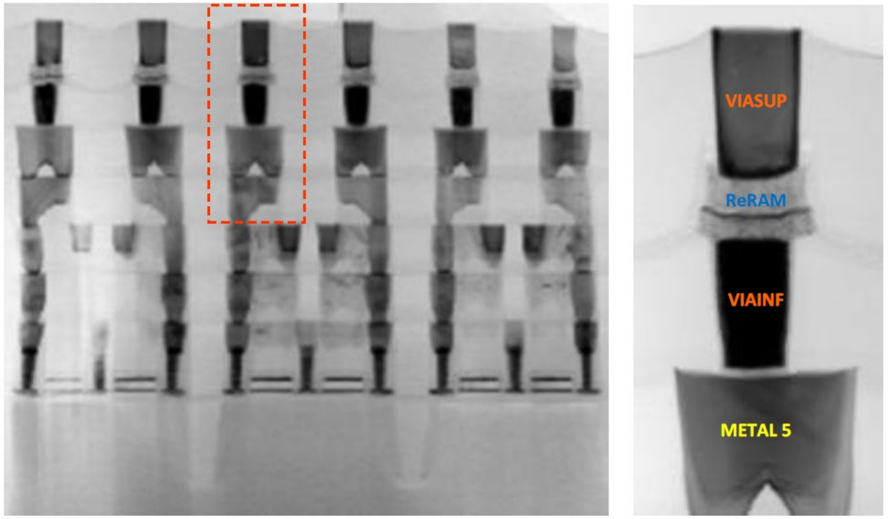


Margin & Error Rate

Example of 1Mb Array



Example of ReRAM Integrated in 28 nm FDSOI



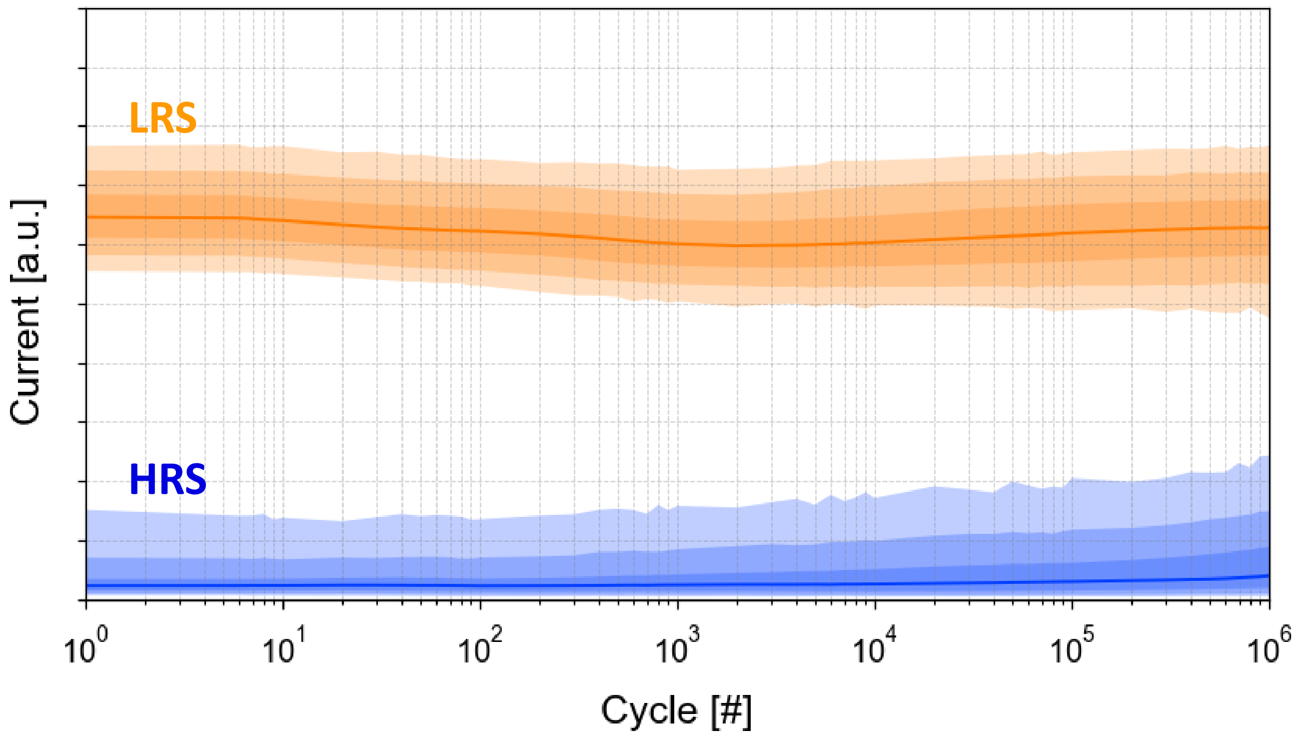
G. Molas, IMW 2022, Weebit [1]

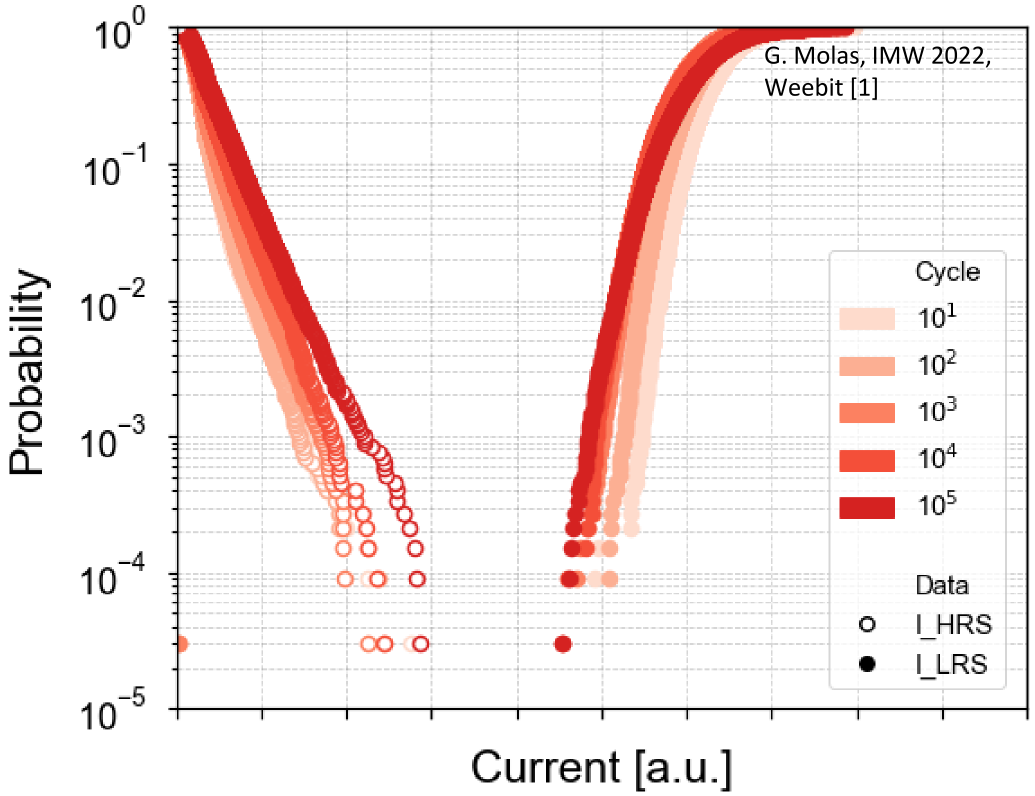
- Low and High resistive states are clearly separated, showing no fails in 1Mb array

Window Margin & Endurance

Example ReRAM endurance in 28 nm node

G. Molas, IMW 2022, Weebit [1]

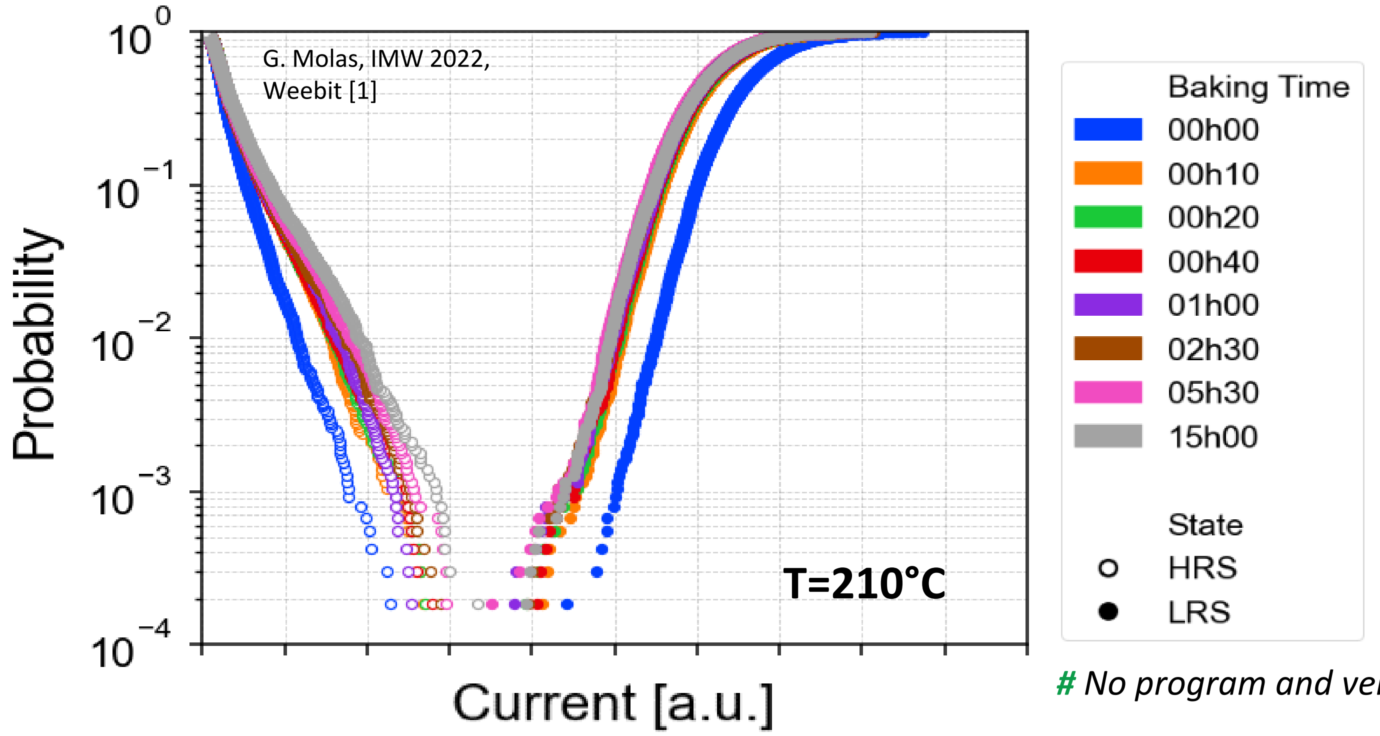




- Typical ReRAM endurance in ~10^5-10^6 cycles

Retention

Example ReRAM retention in 28 nm node



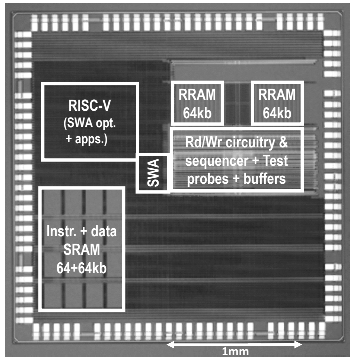
G. Molas, IMW 2022, Weebit [1]

 # No program and verify algorithm was used.

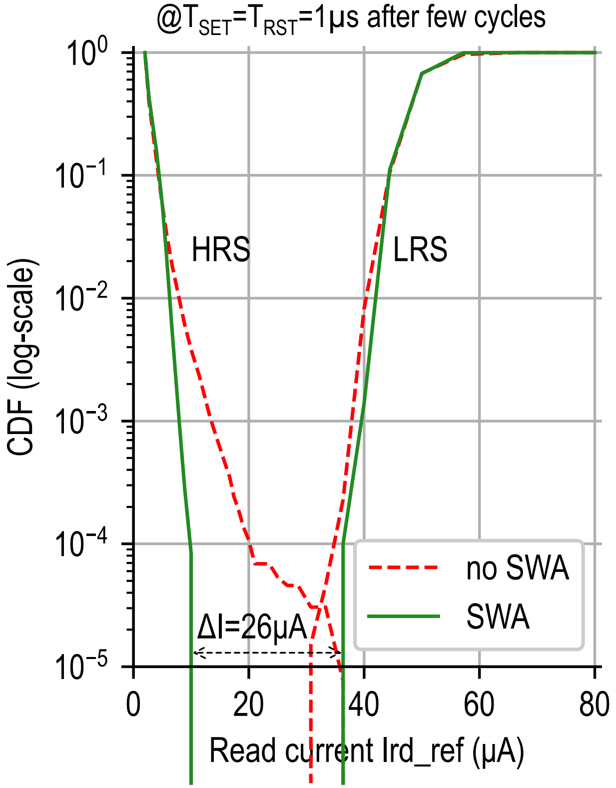
- Slightly small current drift after 15h at 210°C, showing solid cell reliability demonstration
- Most of the drift occurs during a short period, due to cell relaxation / filament stabilization

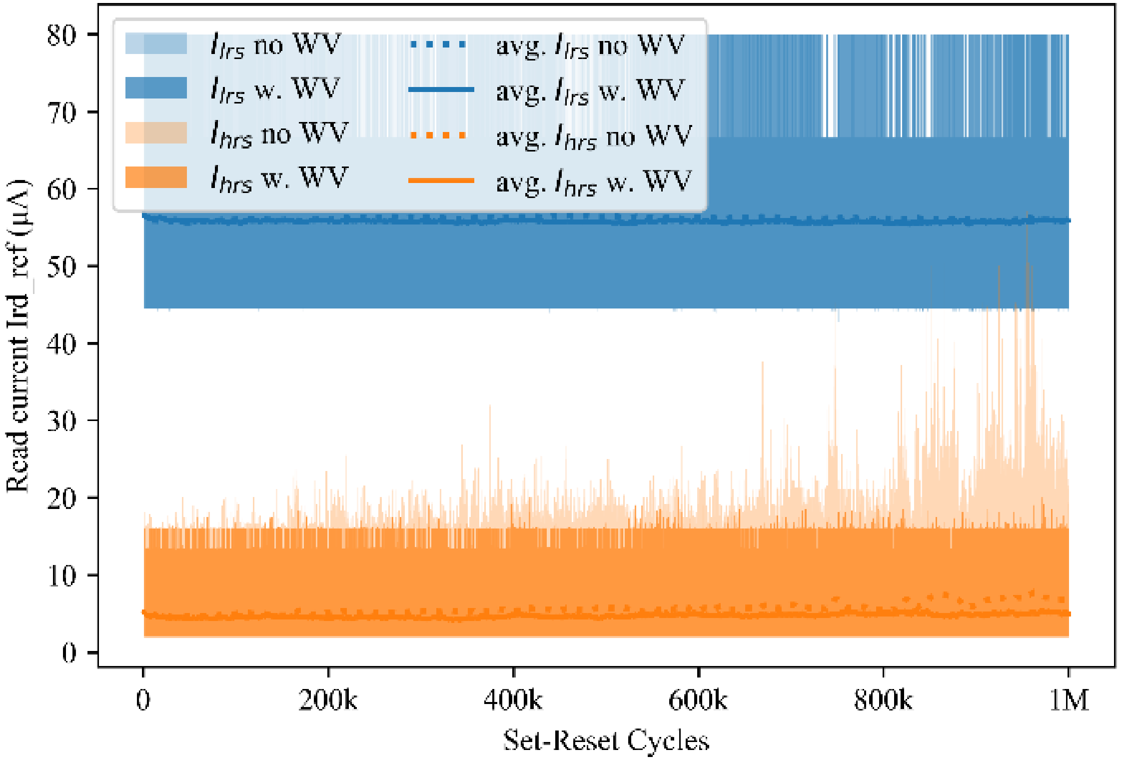
Prog. & Verify on IP Module

B. Giraud, IMW 2023
Leti – Weebit [2]



Read Margin with optimal SWA parameters

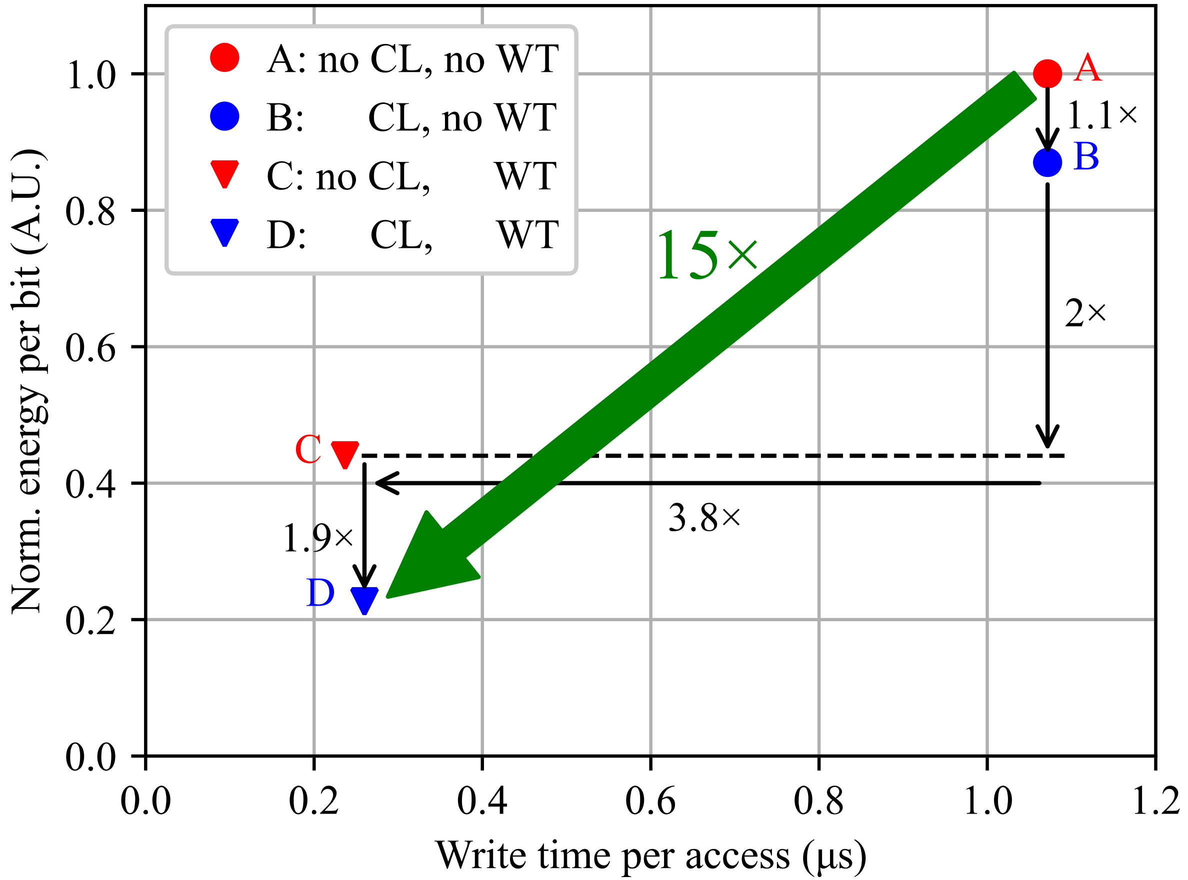




- Margin and endurance measured on 130nm Memory IP Module
- Design options including prog. and verify allow to insure no fail on 128kb arrays and 10^6 endurance

Power Consumption

- CL = current limiter
- WT= write termination



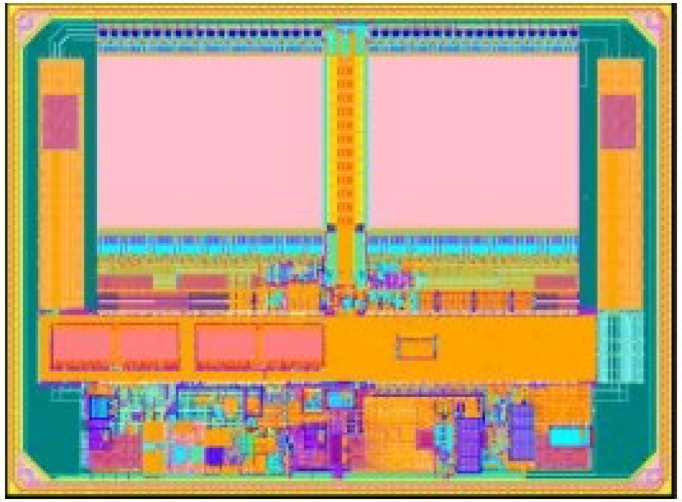
B. Giraud, IMW 2023
Leti – Weebit [2]

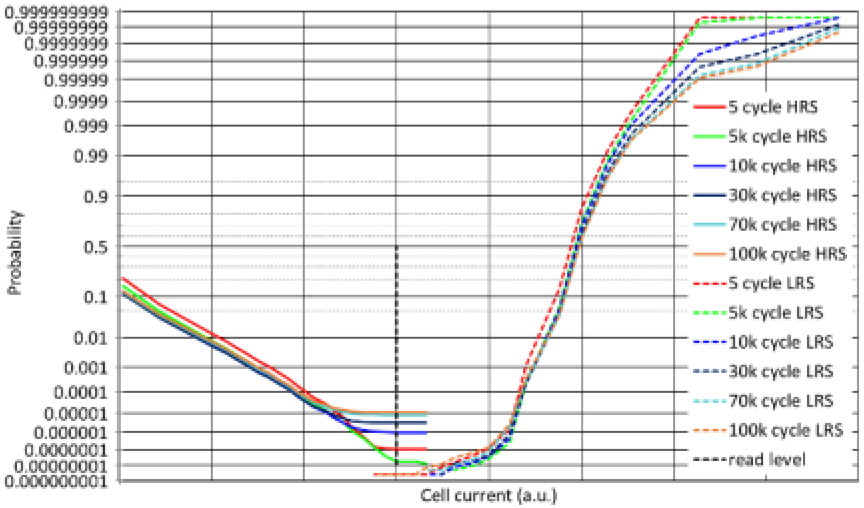
❖ CL and WT help improve the power consumption

ReRAM State-of-the-Art

C. Peters, IMW 2022
Infineon [3]

- RRAM in 28 nm advanced logic
- Thanks to various programming algorithms the failure rate can be brought down below the ECC capabilities
- Demonstrated 15 [year] @ 85 [°C] retention with accelerated experiments





ReRAM Fits Various App Requirements

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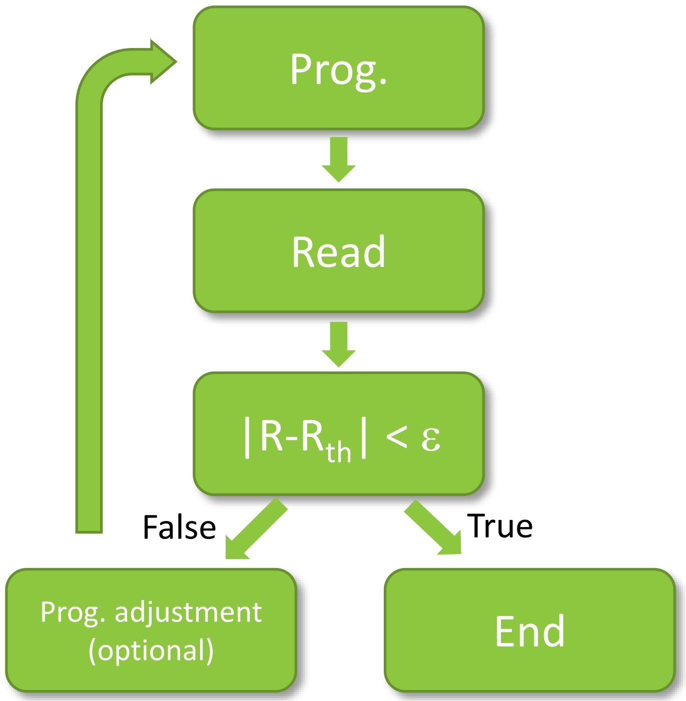
	Mixed-Signal / Power Mgmt	IoT / MCUs	Edge AI	Automotive	Aerospace & Defense
Back-end-of-line tech for easy analog integration	✓				
Cost-efficiency	✓	✓	✓	✓	
Ultra-low power consumption	✓	✓	✓		
Robustness in high temp / extreme environments	✓	✓		✓	✓
Scaling advantage at 28nm and below		✓	✓	✓	
High Endurance		✓		✓	✓
Small footprint to store very large arrays			✓	✓	
Longevity		✓		✓	✓
Roadmap to neuromorphic computing			✓		

Pro/Cons of ReRAM for Neuromorphic

Pro	Cons → Opportunity?
Nonvolatile behavior	The limited precision of the programmed conductance
Good scaling	Conductance drift
Integration with the back end of the line (BEOL)	Relaxation(Time, Temp)
Low current consumption	Random telegraph noise (RTN)
Ability of multilevel cell (MLC) operation	

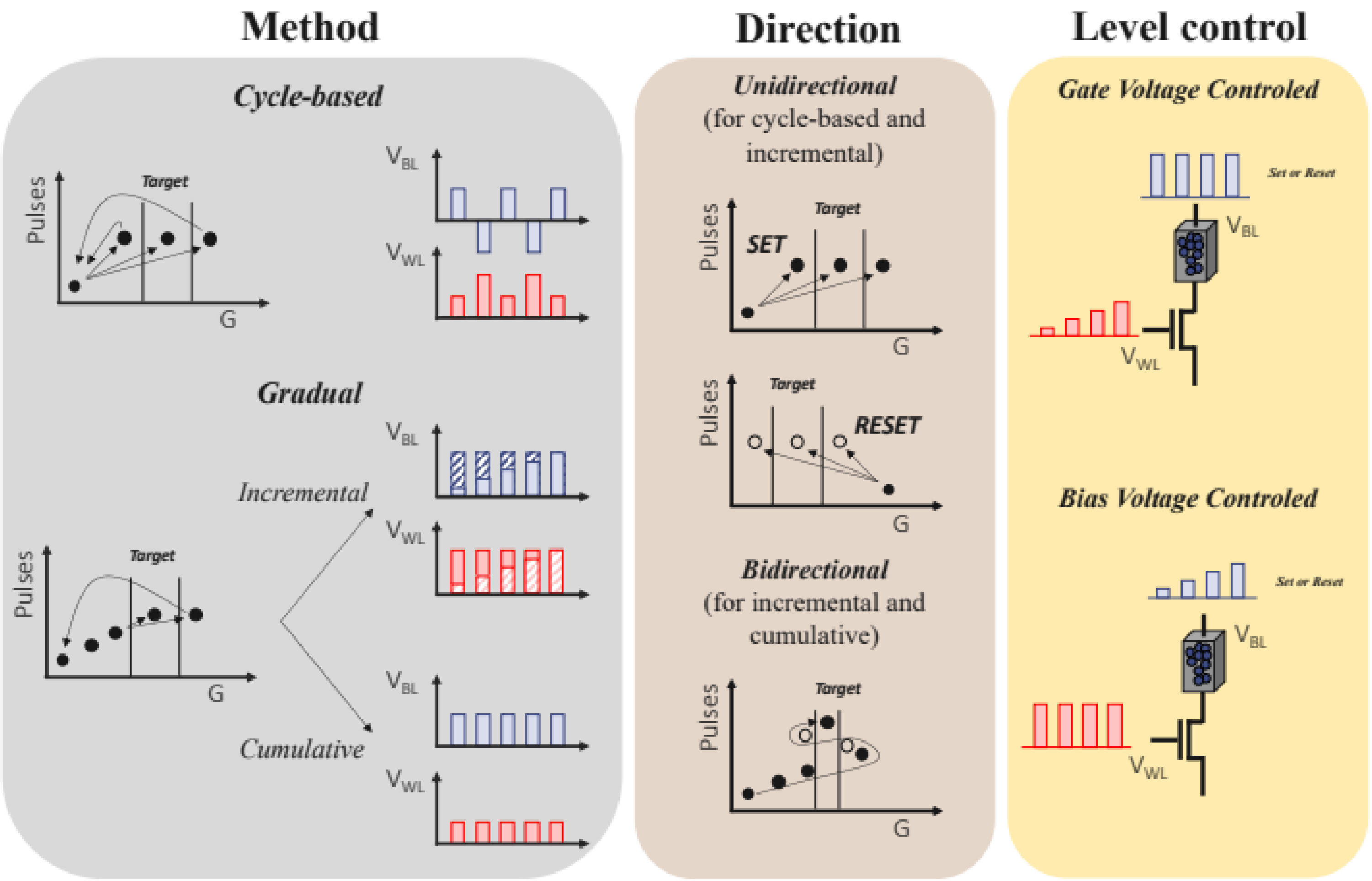
- Limitations of ReRAM can be taken advantage of in bio-inspired NN that are more resilient to variability

Program & Verify for Better Resistance Control



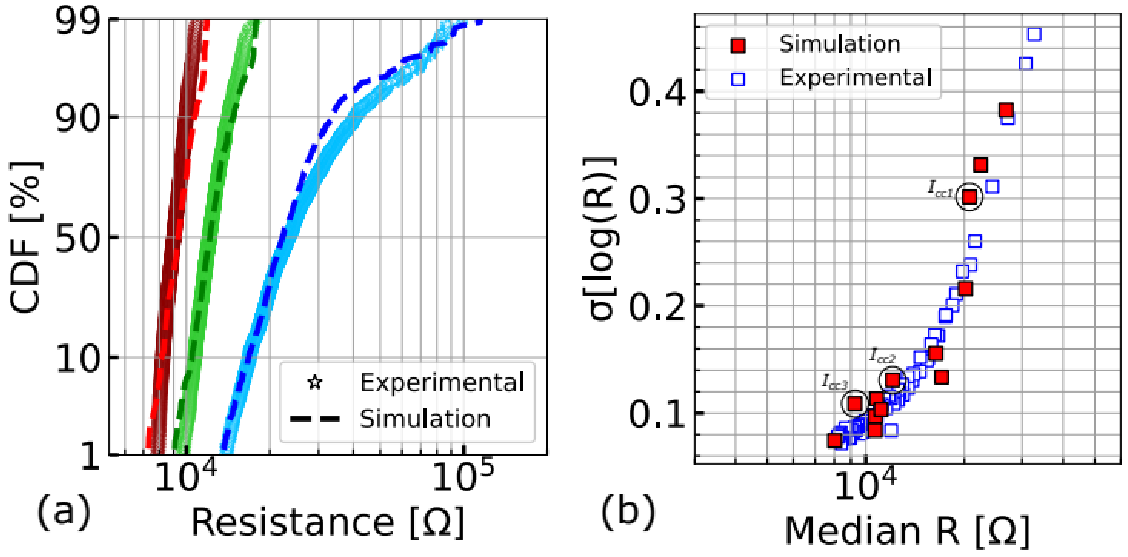
- Many algorithms exist in the literature, playing with current, voltage, sequence… Aiming at resistance distribution control
- When the resistance after programming is not in the expected range, the cell is reprogrammed

P&V Algorithms

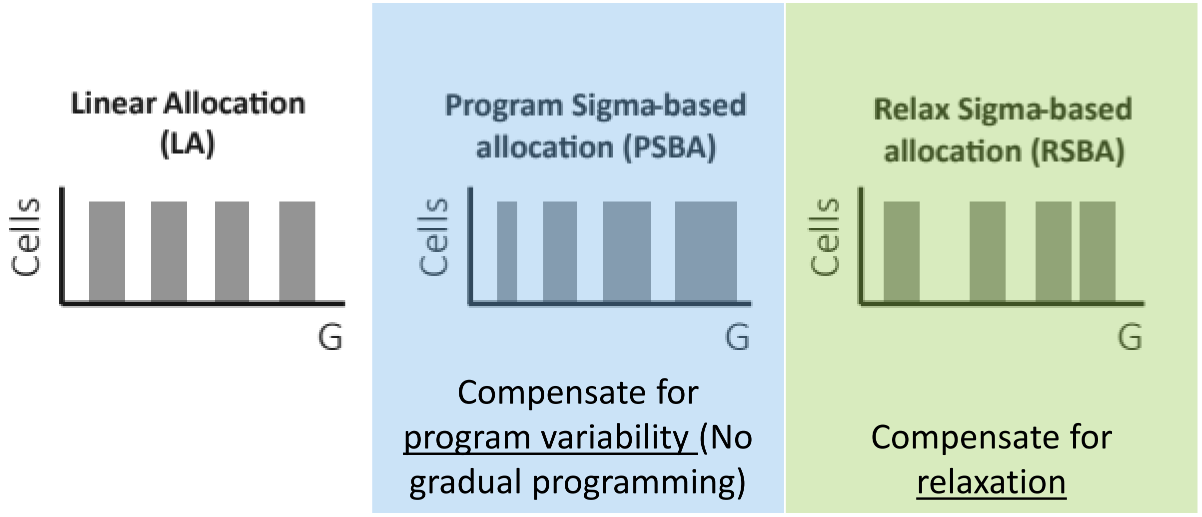


ML Spacing

L. Reganaz PSSA 2022, Leti [4]



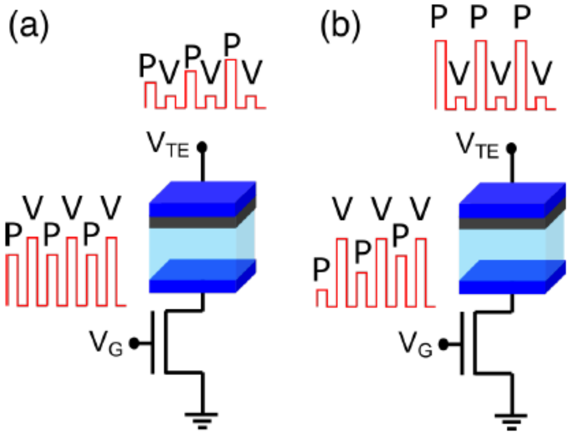
E. Esmanhotto IEDM 2020, Leti [5]



Resistance levels = f(Program variability, relaxation) →ML spacing need to be adapted

State-of-the-Art Algorithms

↘ Variability



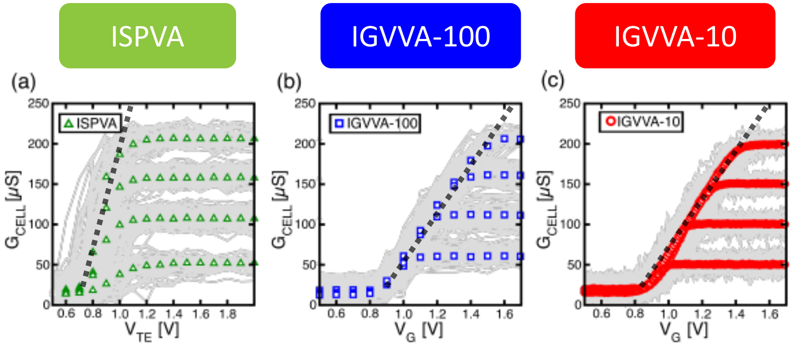
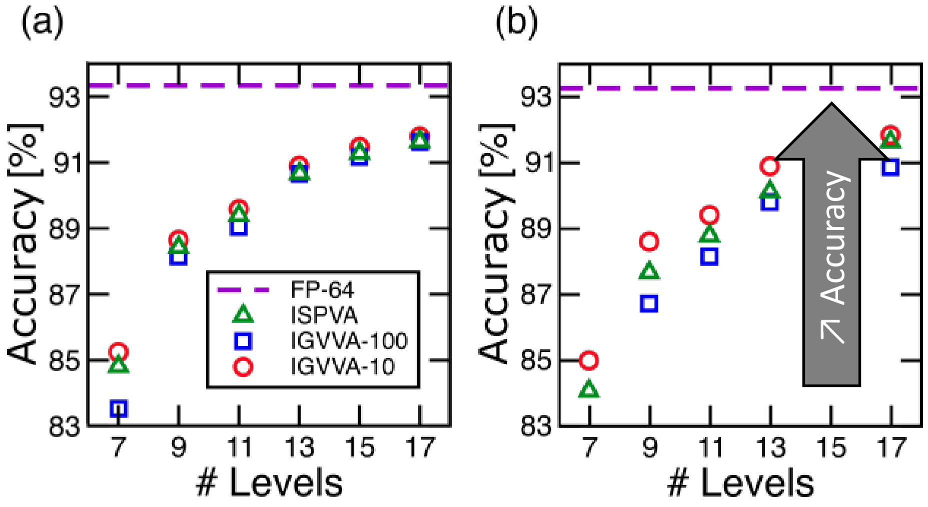


Fig. 3. Measured read current during verify phase V of Program and Verify algorithms. Four different LRS are analyzed. (a) ISPVA is performed with steps of 100mV between programming pulses; IGVVA, instead, is designed with (b) 100mV and (c) 10mV voltage steps respectively. (In grey) Device to devices measurements and (colored) median value. [16]

A. Glukhov, IRPS 2022, Politecnico di Milano [6]

- Algorithms allow to attain various resistance states
  - ISPVA: staircase (SC) on the V_{TE} (top electrode)
  - IGVVA: staircase (SC) on the V_G (selector gate that defines the compliance current)
- Conductance variability of each level can be tuned with the algorithm
- Variability impact on performance
  - Accuracy does improve when variability is reduced but gain is limited thanks to the intrinsic robustness of the Neural Network model

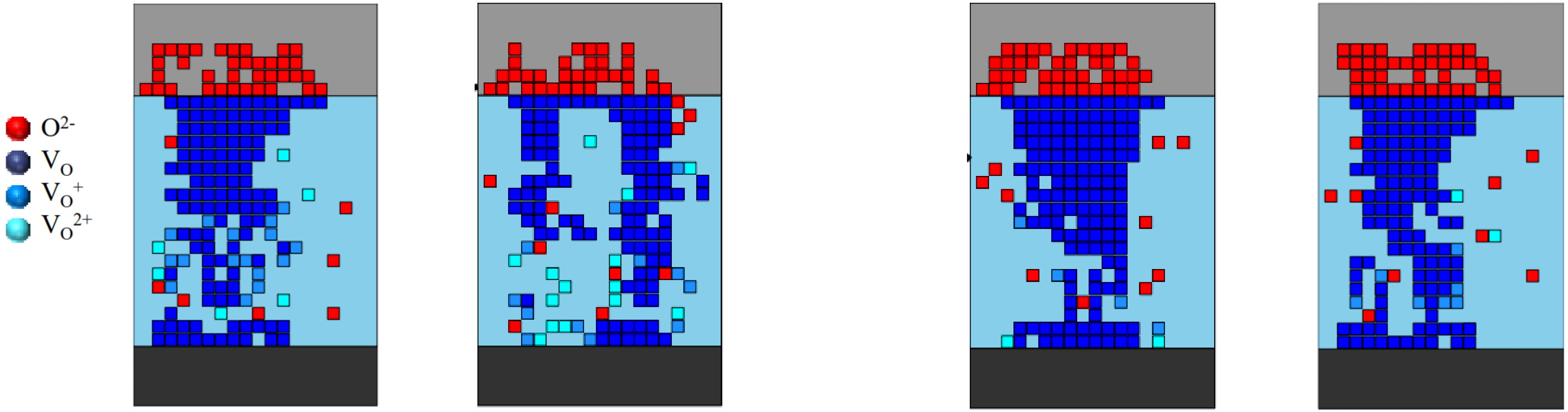


Algorithm for Reduced Fluctuations

- Filament post programming (KMC simulations)

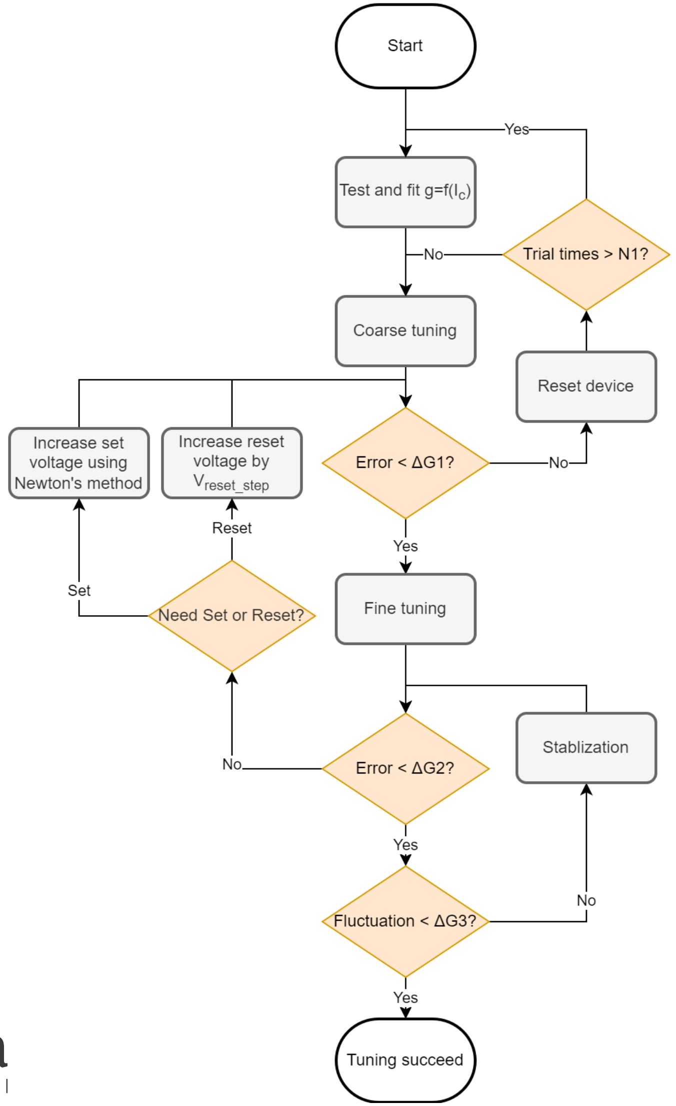
L. Reganaz, SSDM 2023
Leti – Weebit [10]

Unstable **Stable**

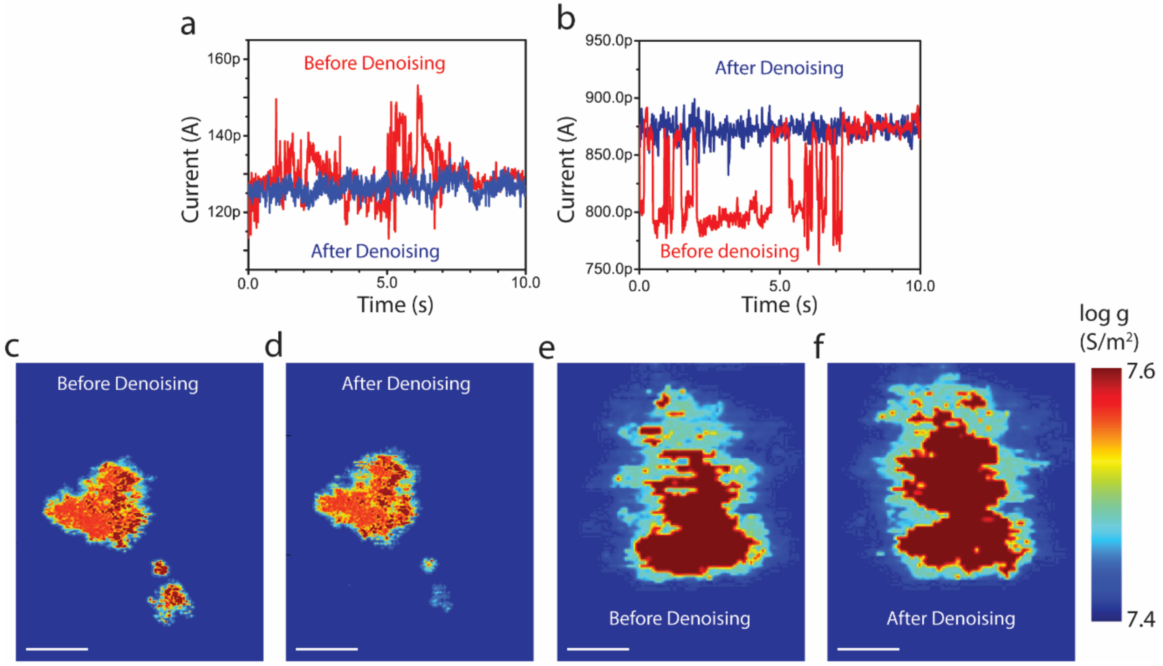


- In some cases, ReRAM may be unstable after programming due to too much VO^{2+}/VO^{+} mobile vacancies around the VO filament
- Prog and verify algorithm basically allows to reach a more stable filament configuration, reducing post programming fluctuations

Noise Mitigation

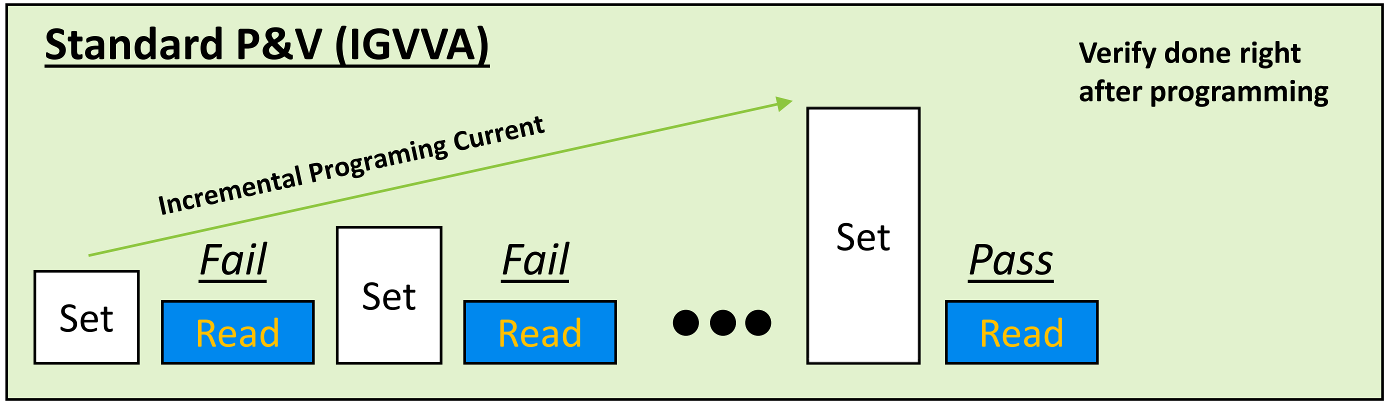


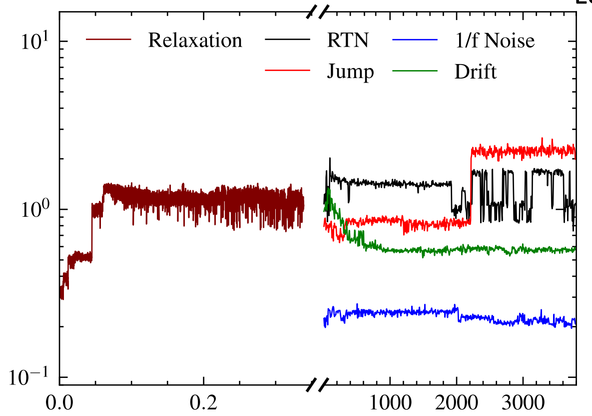
M.Rao, Nature, 2023, University of Massachusetts [11]

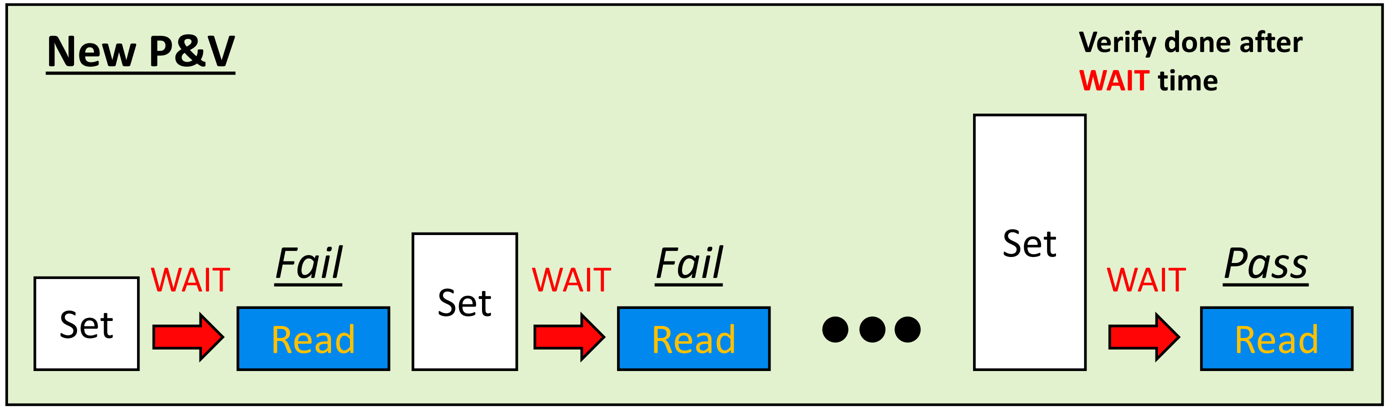


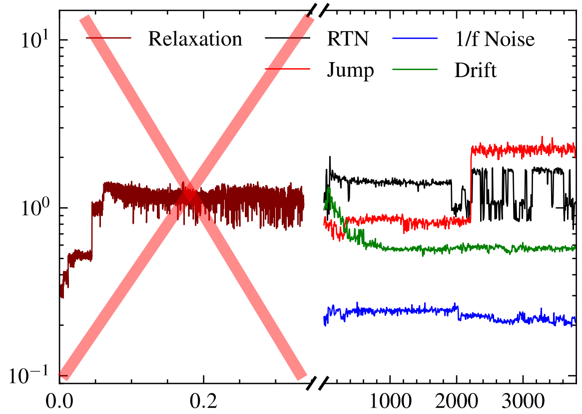
Algorithms with Delay to Improve ML – Concept

L. Reganaz, IRPS 2023
Leti – Weebit [9]





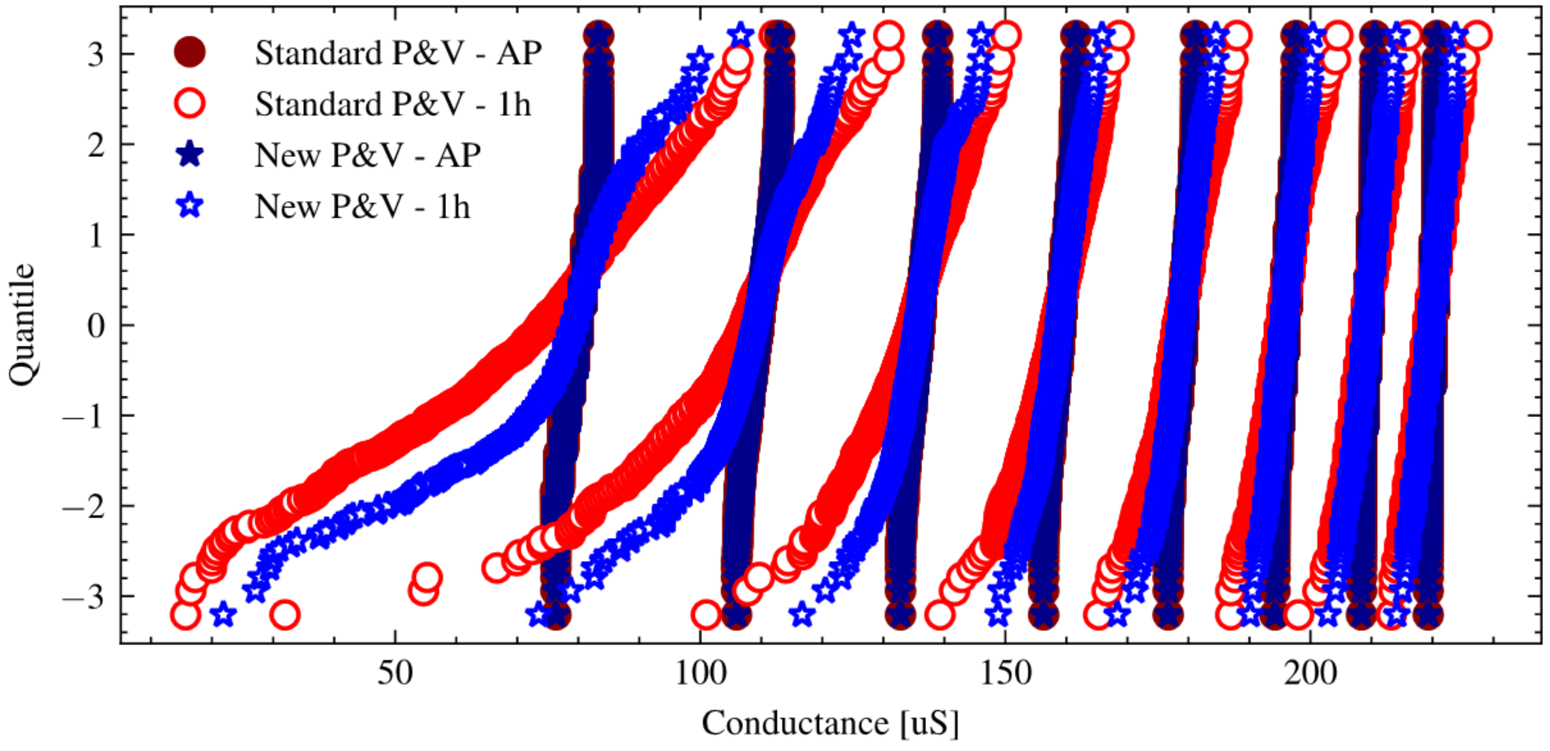


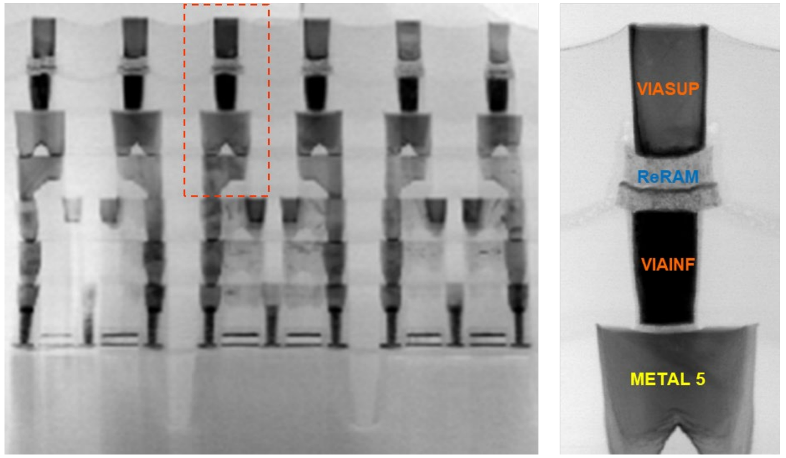


Part of variability that can be removed

Algorithms with Delay to Improve ML – Experimental Results

L. Reganaz, SSDM 2023
Leti – Weebit [10]



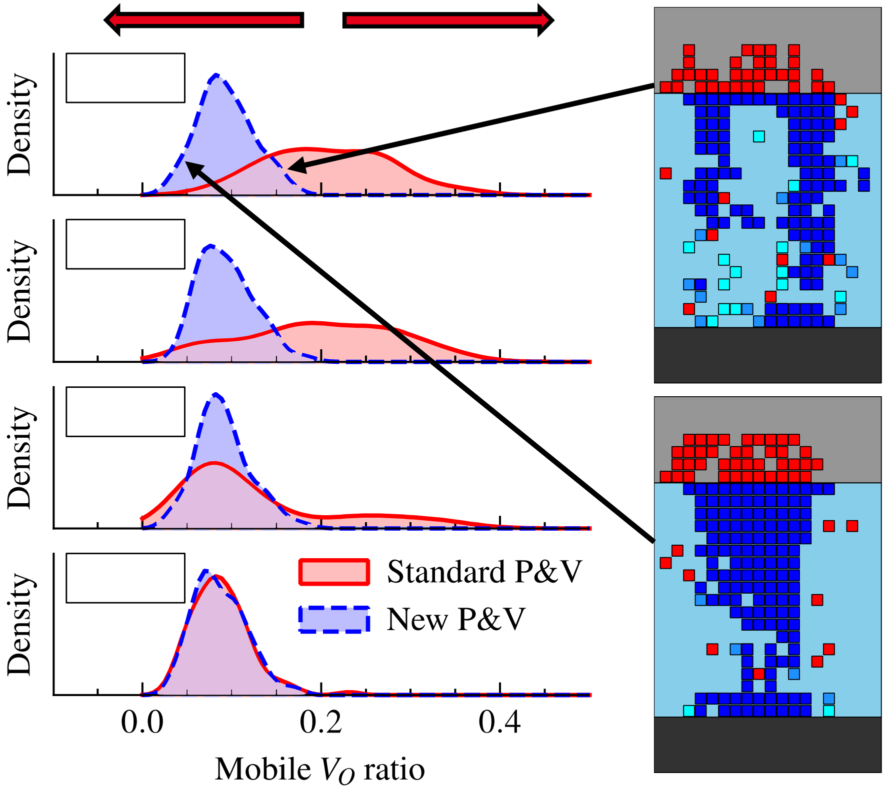


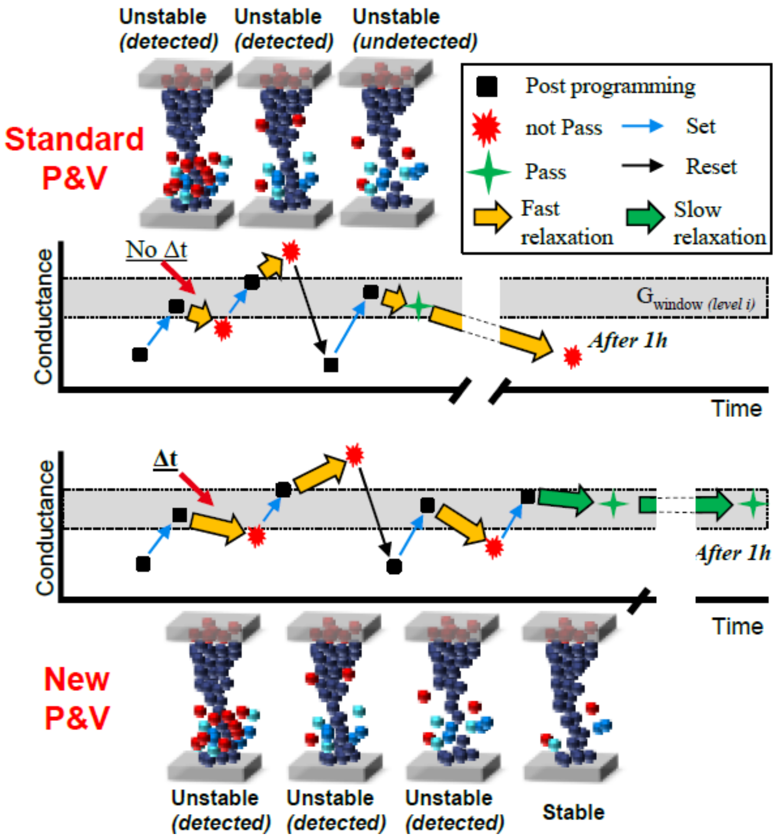
- 16kb 1T1R array in 28nm FDSOI BEOL

- By adding 1min verification delay to standard P&V, we can achieve better MLC stability
- What's the physical explanation

Conductance Instability from Highly Mobile V_O

L. Reganaz, SSDM 2023
Leti – Weebit [10]

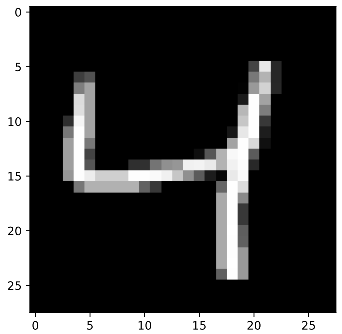




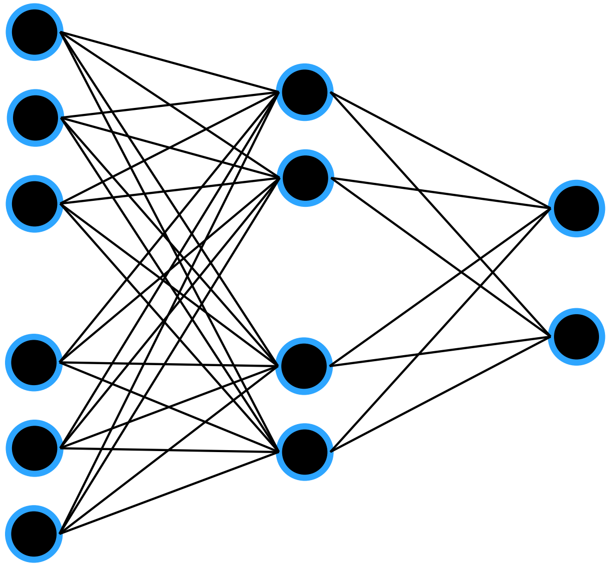
- New P&V algo allows to reduce the number of high mobile V_O because thanks to the waiting time we can "select" only the conductive filament with fewer high mobile V_O.

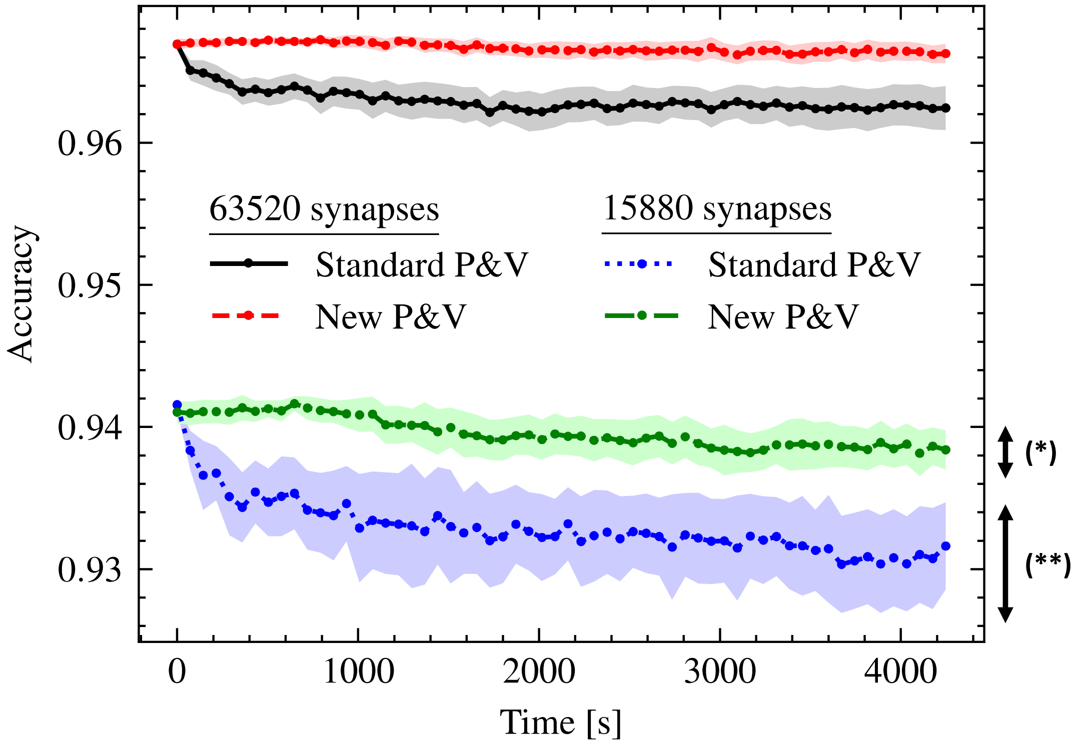
Impact of Improved Algorithms on NN Circuit Performance

L. Reganaz, SSDM 2023
Leti – Weebit [10]



- MNIST dataset

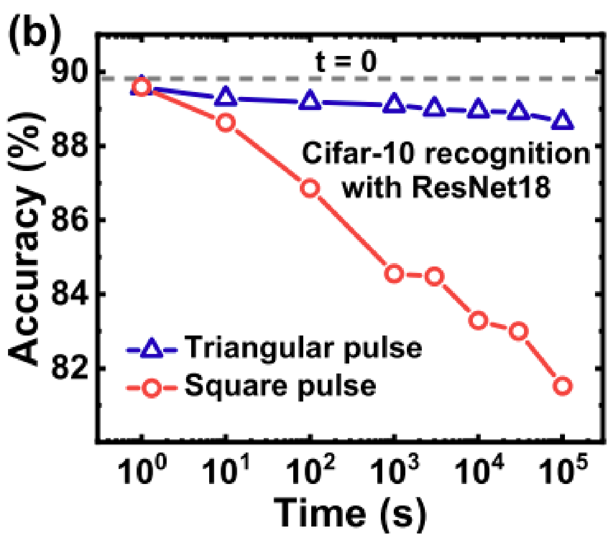


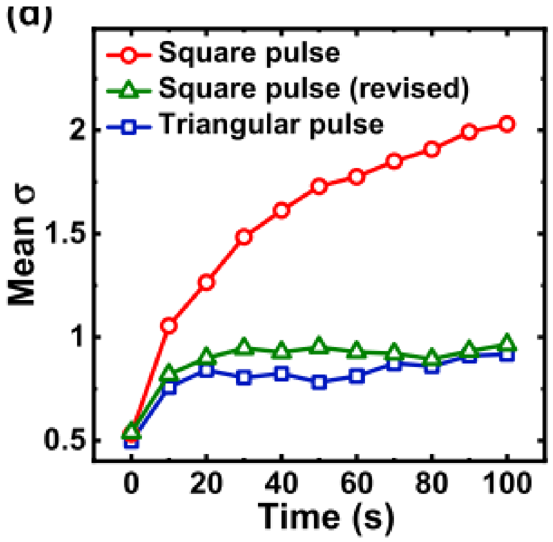


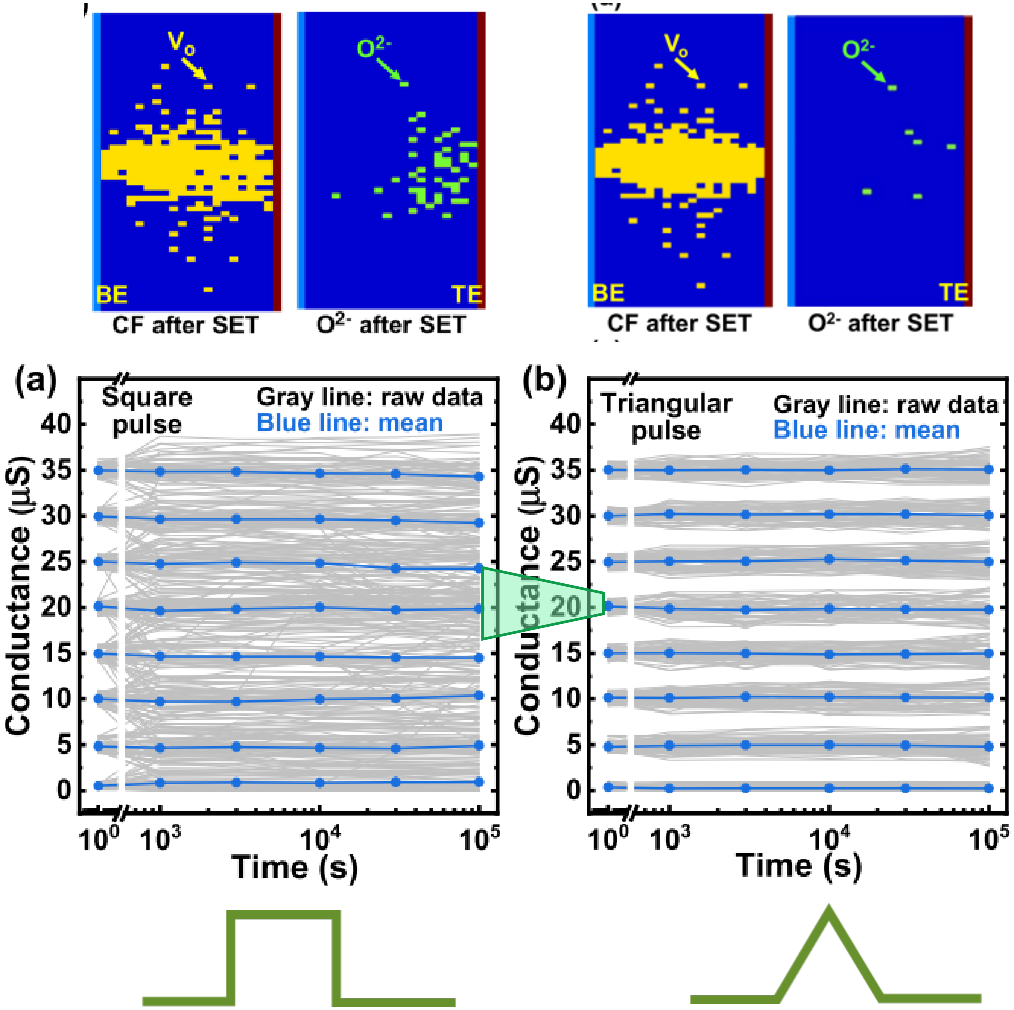
- Better NN inference accuracy stability through time
- Better weight transfer precision (=lower accuracy deviation on repeated simulations, * vs **)

Impact of Pulse Shape on Variability

Y. Feng et al., EDL 2021, IME [12]







- Triangular pulse shape reduces Vo after set operation therefore improving conductive filament stability
- Improved accuracy of NN

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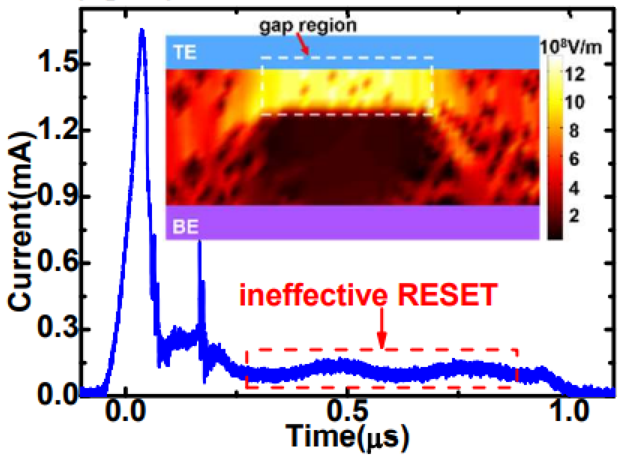
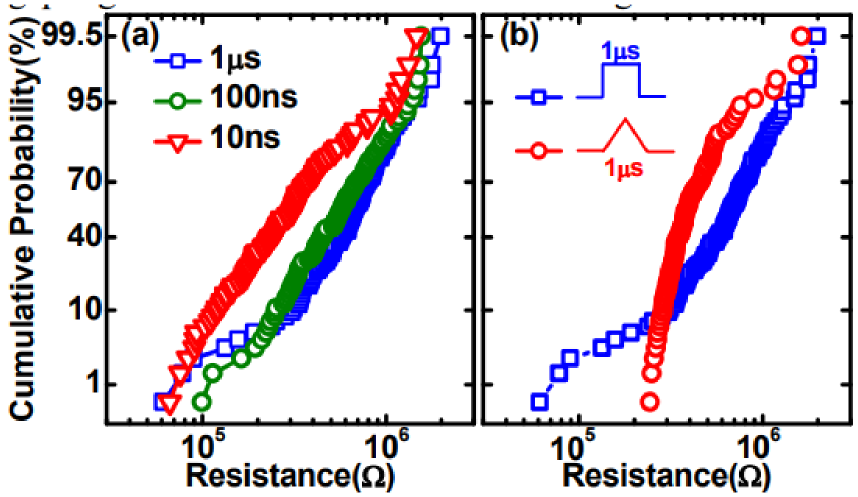
Impact of Pulse Shape on HRS

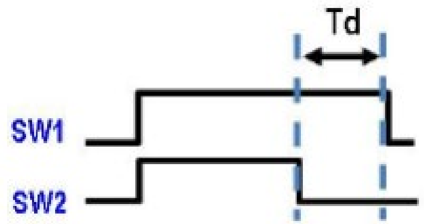
P. Huang et al., IEDM 2017, IME [13]

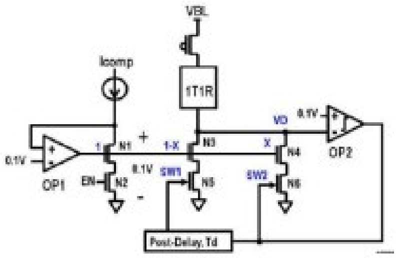


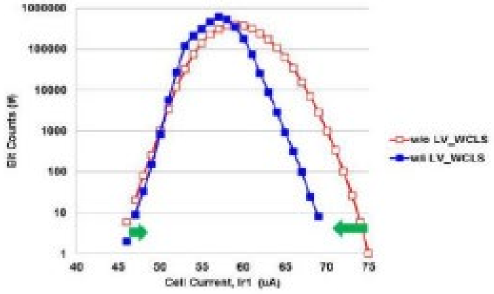


- Triangular pulse shape reduces Vo by avoiding the "ineffective Reset" in the reset operation therefore improving HRS tails

Impact of Program Strategy on LRS







Chou et al., ISSCC 2018, TSMC [14]

SW2 (standard write termination):

- Pulse is interrupted once switch is detected

SW1 (improved write termination):

- Pulse is interrupted with a given delay after switch is detected

- Adding a delay when doing the write termination improves conductive filament stability

ReRAM for Neuromorphic

ReRAM perfectly fits in brain inspired systems, in a timely manner:

Embedded, Edge AI

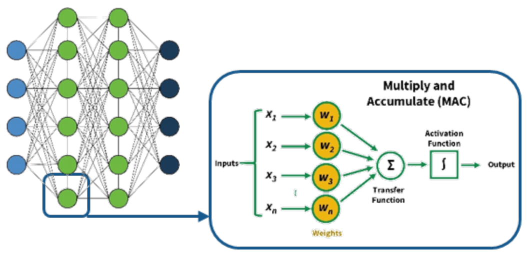
Short Term

Mid Term

Longer Term

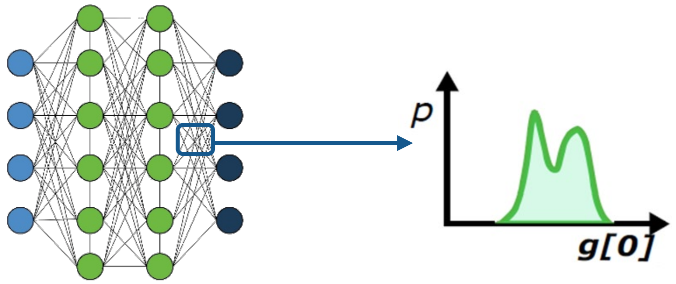
New Concepts

In-Memory Compute:
AI and ML



synaptic weight
storage

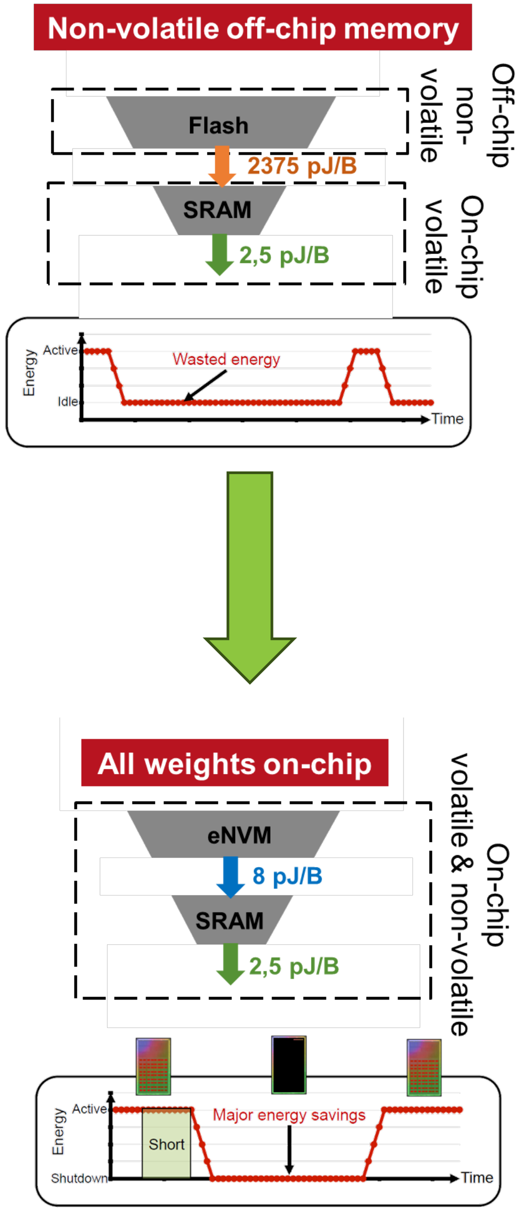
IMC



Bayesian

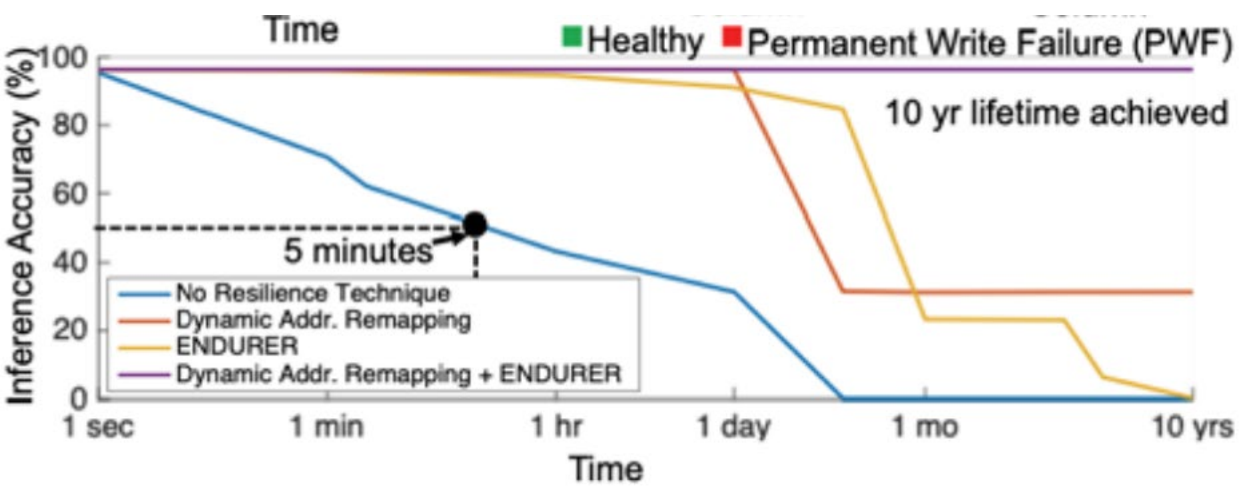
Usage of ReRAM for AI / IMC

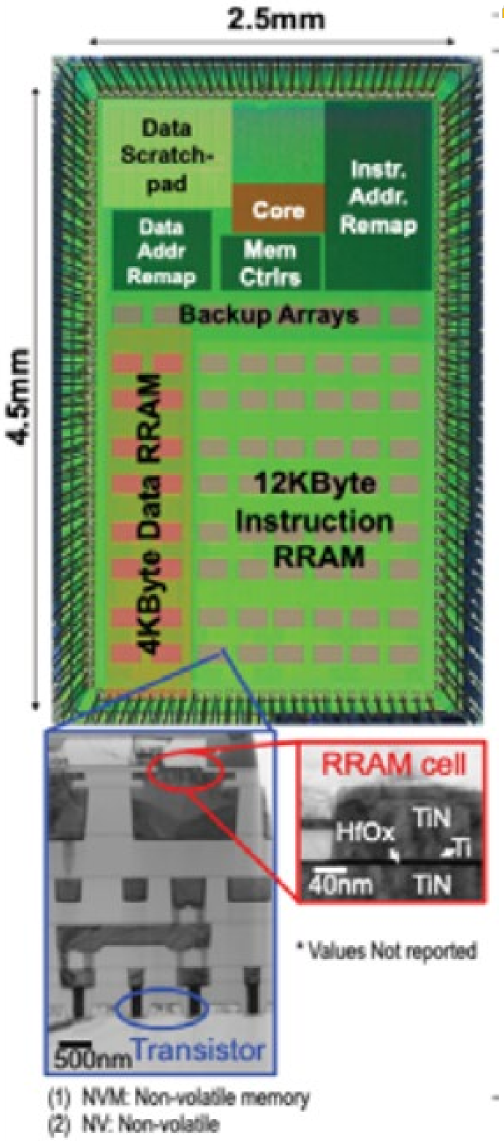
- **Short-term**: The first opportunity for ReRAM in edge AI as embedded memory, 10MB-100MB
  - Bringing NVM closer to the computing engine
  - Smaller than SRAM (>2x) and non-volatile
  - Endurance can handle infrequent update of weights
  - Moving from 1b into multi-level as AI is less sensitive to overlapping bit (see diagram of 9 levels)



RRAM in DNN

F. Wu, ISSCC 2019, Stanford [15]





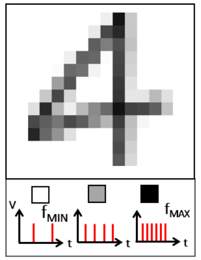
- ReRAM stores synaptic weights of DNN (5 resistance levels)
- ReRAM gives an advantage in terms latency wrt flash and lower energy features

Fully Integrated Spiking Neural Network Using Weebit ReRAM as Synaptic Device

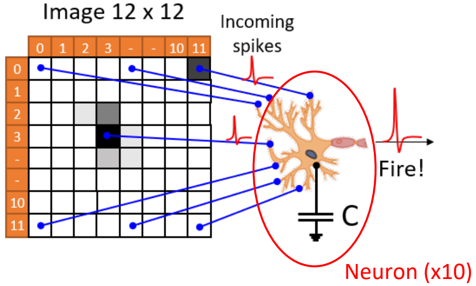
Proof-of-concept SNN performing recognition of MNIST digits

(Off-line learning)

Digit drawn on tablet application



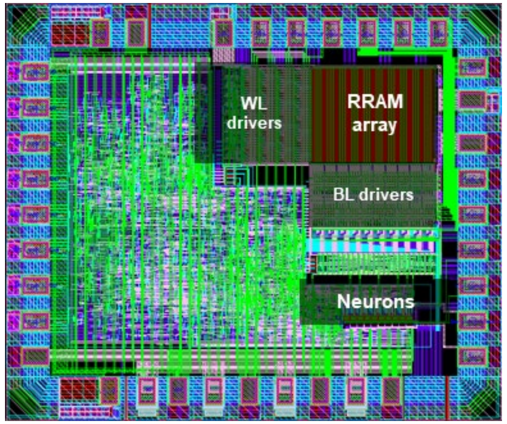
256 grey levels Converted in spike frequency



Fully-connected, single-layer perceptron

A. Regev, AICAS 2020, Weebit [16]

- Synapse design
  - 8 ReRAM (4x excitatory / 4x inhibitory)
- System performance
  - 160 spikes, on average, to recognize an image
  - Measured energy consumption: 180 pJ/syn. ev. (3 pJ at synaptic level)
  - 82% of accuracy obtained – to be compared with the maximum theoretical accuracy 88% (topology-limited)

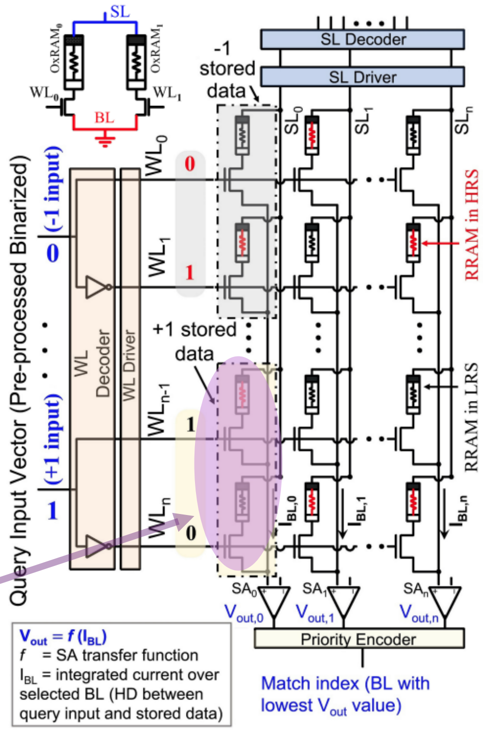


System details

CMOS process	130 nm
Cu interconnects	5
RRAM number	11,5k
RRAM configuration	1T-1R
Chip size	1.8 mm^2

In-Memory Computing for AI Similarity Search

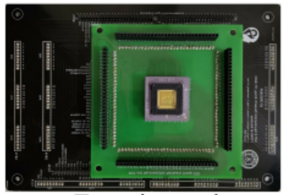
- CAMs (Content Addressable Memories) offer a powerful in-memory computing paradigm for efficient parallel-search and pattern-matching applications
- Query match/mismatch based on the current readout (*)
- 2T2R bitcell used to implement XOR operation
- 2T2R allows for great resilience to D2D (Device-to-Device)/C2C (Cycle-to-Cycle) variability (for both LRS and HRS)



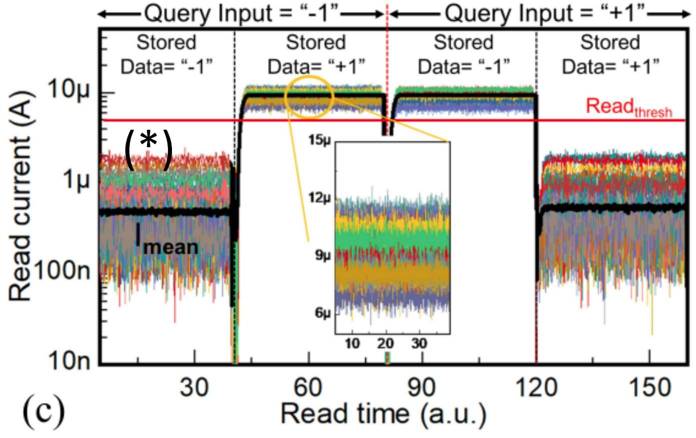
S. K. Kingra, TCS II, 2023 [17]

Query Input	Stored Data	Bitwise XOR	Device Read
-1	-1	-1	HRS
-1	+1	+1	LRS
+1	-1	+1	LRS
+1	+1	-1	HRS

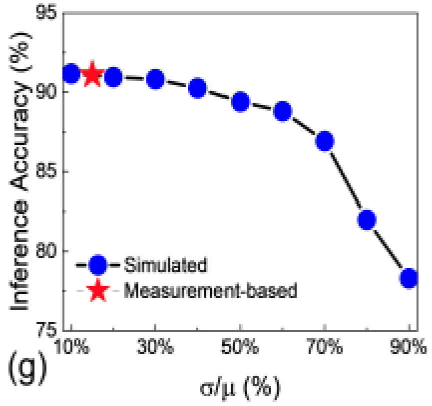
(a) 2T-2R XOR Truth Table



(b) Experimental Testbench



(c)

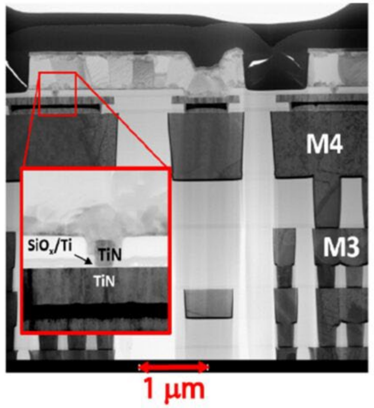


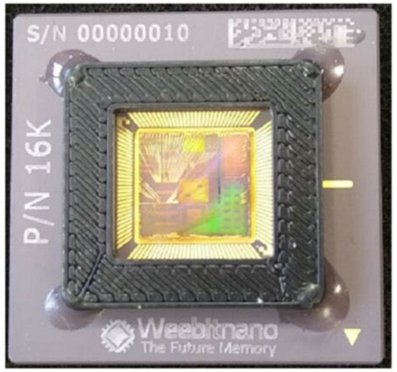
(g)

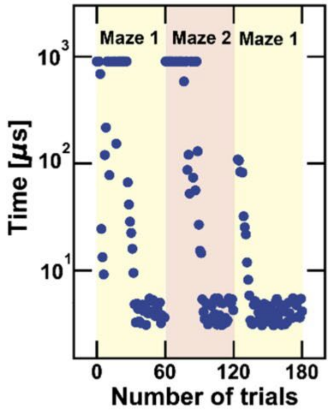
Bio-inspired Neural Network

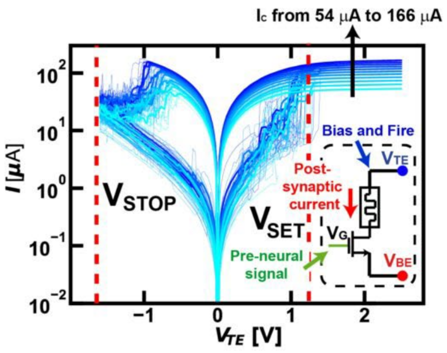
- **Plasticity**
  - Why is it important? Allows to adapt more quickly to changing environment
  - How to achieve it? ReRAM conductance can be changed with few electrical parameters

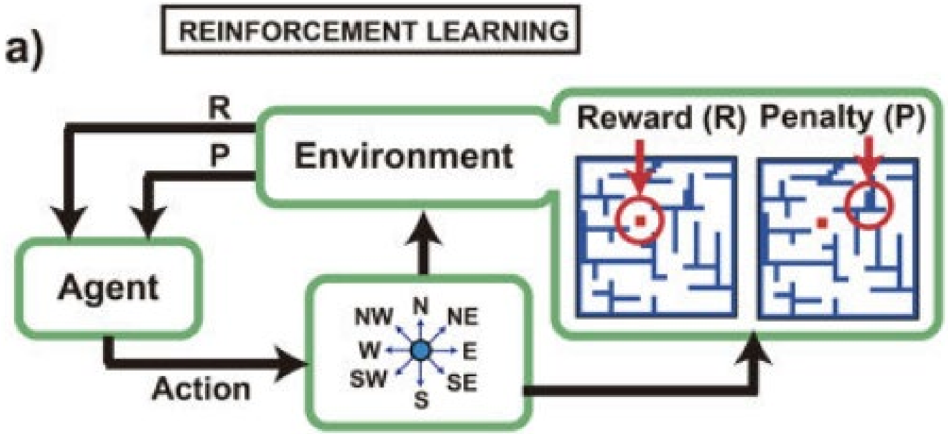
- **Weebit's ReRAM devices were used**
  - Store information and adjust the strength of connections between neurons
  - Map the internal state of each neuron







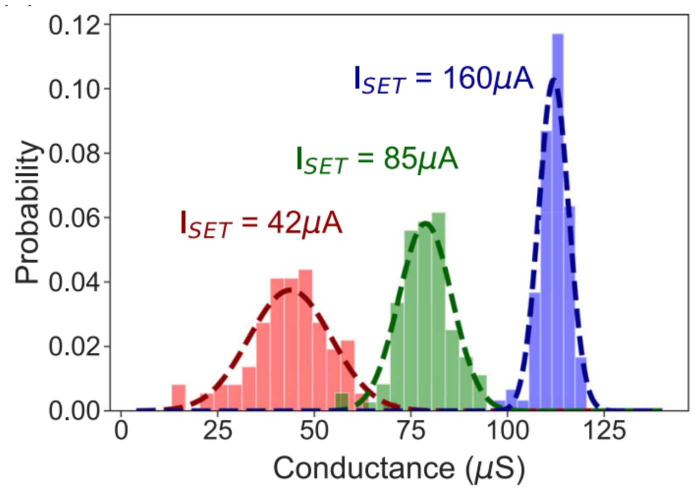




S. Bianchi, Nature, 2023,
Politecnico di Milano [18]

Usage of ReRAM for new concepts

- **<u>Longer term</u>**: new concept such as Bayesian NN



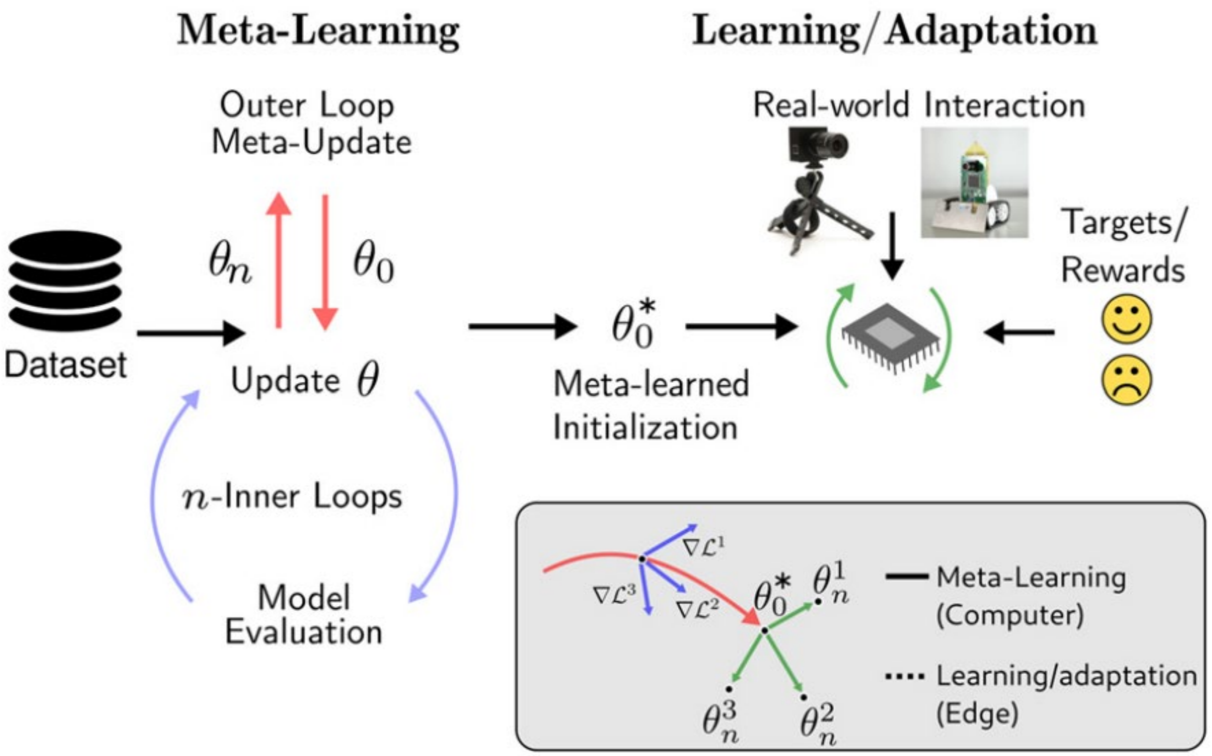
T. Dalgaty *et al.*, Adv. Intell. Syst., 2021 [19]

Concept Idea of In-situ Learning

Model-Agnostic-Meta-Learning

- OFF-chip:
  - MAML model trained on **server** (over than 50k iterations on GPU)
  - Output off-chip:
    - Optimized net parameters
- ON-chip:
  - Standalone ReRAM device able to quickly learn **new tasks**
  - Confidentiality guaranteed between customers and between customer and company

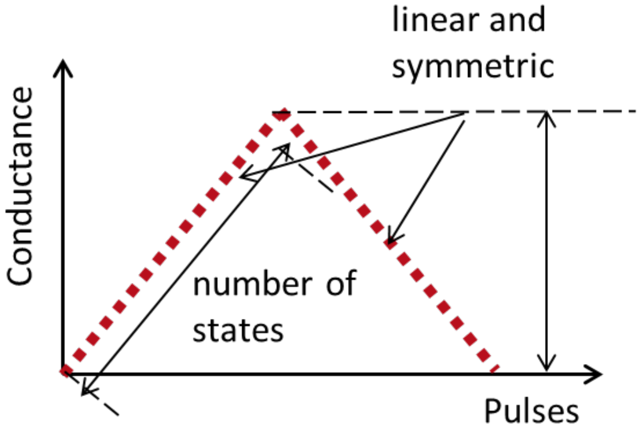
K. Stewart, NCE 2022, UC Irvine [20]



How to Take Advantage of ReRAM Weaknesses

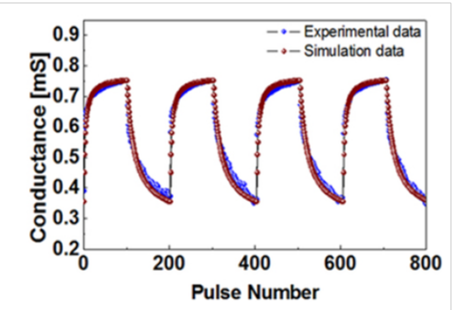
The synapses we would like:

- Linear and symmetric response
- Large number of analog states
- High ON/OFF ratio
- High endurance
- No static or dynamic variability



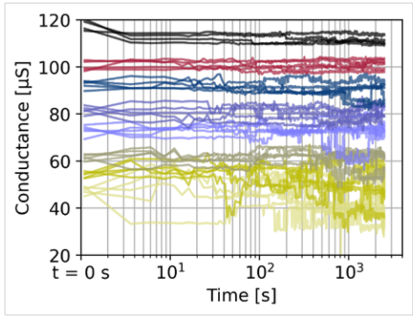
The synapses we have:

Asymmetry / non-linearity



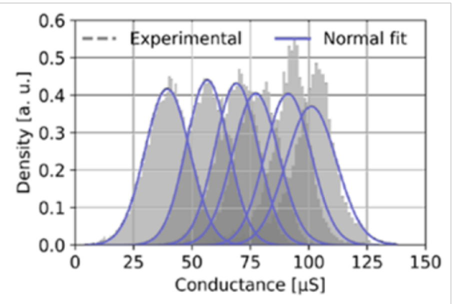
S. Choi *et al.*, Sci. Reports, 2015

Noise / relaxation



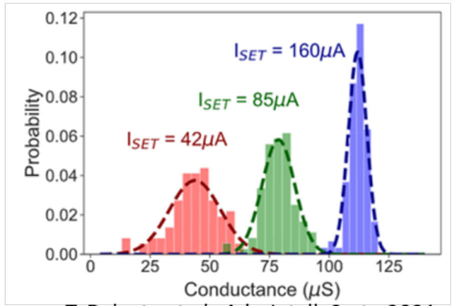
E. Esmanhotto *et al.*, Adv. Intell. Syst., 2022

Device-to-device (D2D)



E. Esmanhotto *et al.*, Adv. Intell. Syst., 2022

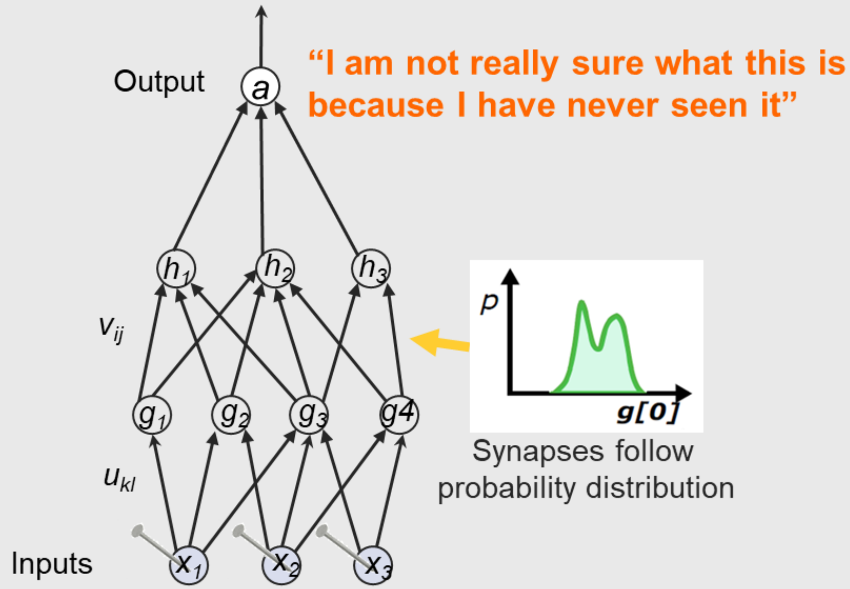
Cycle-to-cycle (C2C)



T. Dalgaty *et al.*, Adv. Intell. Syst., 2021

How to Take Advantage of ReRAM Weaknesses

Bayesian Neural Network

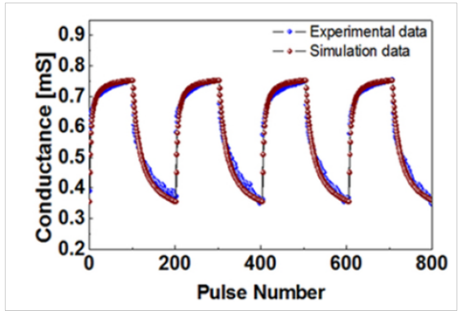


T. Dalgaty, Nature E. 2021, Leti [21]

RRAM intrinsic variability naturally produces Bayesian Neural Network

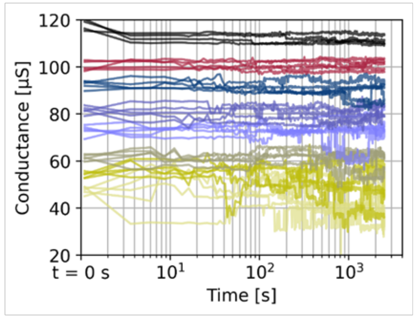
The synapses we have:

Asymmetry / non-linearity



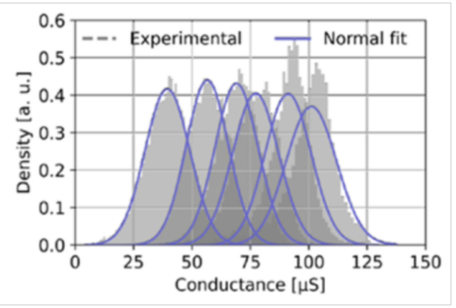
S. Choi *et al.*, Sci. Reports, 2015

Noise / relaxation



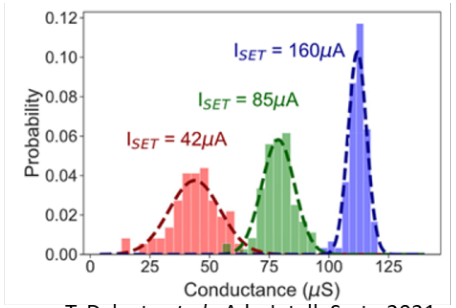
E. Esmanhotto *et al.*, Adv. Intell. Syst., 2022

Device-to-device (D2D)



E. Esmanhotto *et al.*, Adv. Intell. Syst., 2022

Cycle-to-cycle (C2C)



T. Dalgaty *et al.*, Adv. Intell. Syst., 2021

Some models require randomness to compute based on probabilities

Stop fighting non-idealities

Conclusions

- **Neuromorphic applications are a good match for ReRAM thanks to their:**
  - Cost-efficiency
  - Ultra-low power consumption
  - Scaling advantage at 28nm and below
  - Small footprint to store very large arrays
  - Analog behavior
- **Specific stack/operating schemes are required when ReRAM is used for AI applications:**
  - Accurate multi-level
  - Relaxation mitigation
  - Stable states (Retention)
- **ReRAM in Neuromorphic could follow these steps:**
  - Embedded memory for synaptic weight storage (faster than flash, non-volatile, …)
  - In memory computing (analog behavior required, in crossbar arrays)
  - New concepts (taking advantage of ReRAM intrinsic variability for noise resilient systems, e.g., Bayesian)

Acknowledgements

- Elisa Vianello, Louis Hutin and Lucas Reganaz, CEA Leti, Grenoble
- Bastien Giraud, CEA LIST, Grenoble
- Daniele Lelmini, Politecnico di Milano
- Manan Suri, IIT, New Delhi
- Damien Deleruyelle, INL, Lyon
- Quentin Rafhay, IMEP, Grenoble

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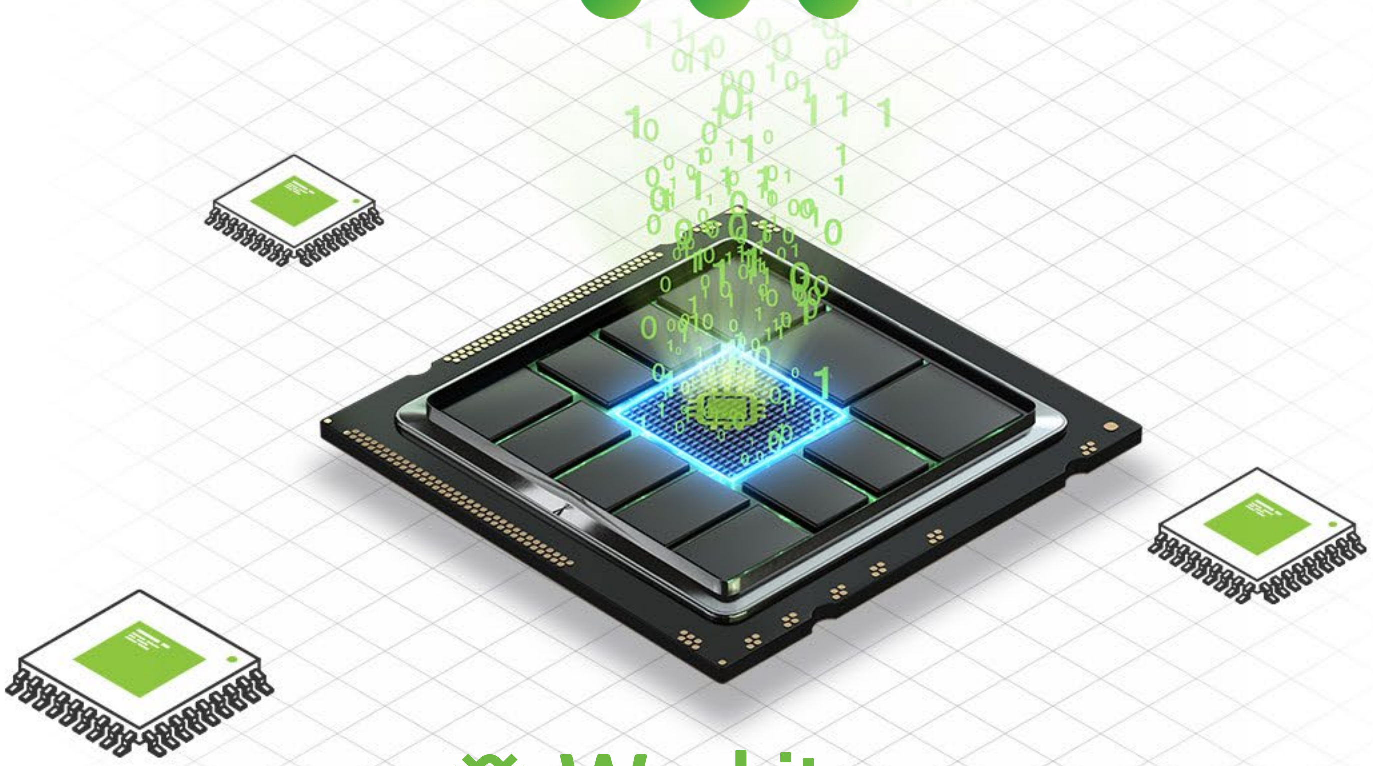

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